The New Work of Generative AI

Evidence from 18 Million Duty Statements in US Job Postings

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October 2026

Abstract

Whether generative AI raises or lowers the demand for labour depends in part on the new work it creates. I measure that new work in 18.4 million duty statements from US job postings, 2019–2026, drawn from a new dataset of applicant-tracking-system pages archived by Common Crawl. Duty statements with no close counterpart in early-2019 postings are flagged as new, and a language model separates new tasks from variants of old ones. In the third of occupations that use AI most, duties that describe new tasks rose from under one in every 100 in 2022 to about three and a half in 2026, most of them about AI itself. Working with AI, rather than building it, rose from 4% of AI duties in 2019–23 to 22% in 2026. Beyond tasks that are new to the entire economy, existing tasks are also diffusing across occupations: between 2022 and 2026, postings requesting technical and data tools, such as dashboards, rose about threefold in occupations whose duties did not name them in 2019–20. Postings with AI duties advertise higher pay, though most of the premium is accounted for by the firms and job titles that post them; within a firm and title it is one or two per cent.

JEL codes: J23, J24, O33. **Keywords:** generative AI, new work, job postings, tasks, reinstatement.

1 Introduction

New technologies displace workers from some tasks and create new tasks for them (Autor et al., 2003). Whether a technology raises the demand for labour on net depends on how the displacement compares with the gains in productivity and with the creation of new tasks, which Acemoglu and Restrepo (2018, 2019) call reinstatement. Job postings record an early margin of this second effect: the new tasks that employers ask new hires to do. Historically, new work has appeared where innovations complement the tasks that workers already perform (Lin, 2011; Autor et al., 2024). For generative AI, most of the evidence so far is on the other side of the ledger: which occupations and tasks the technology can perform (Felten et al., 2021; Webb, 2020; Eloundou et al., 2024; Tomlinson et al., 2025), which ones people use it for (Handa et al., 2025; Massenkoff and McCrory, 2026), and whether employment in exposed occupations has fallen (Brynjolfsson et al., 2025a; Hampole

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et al., 2025; Humlum and Vestergaard, 2025). The studies that look for new work count skills new to a firm and occupation in postings (Chen et al., 2024) or ask workers whether chatbots have created new tasks (Humlum and Vestergaard, 2025), rather than reading the duties that employers advertise.

A job posting lists what an employer asks a new hire to do, one duty at a time. Using 1,629,317 US postings captured from 14 applicant tracking systems (ATS) archived by Common Crawl between 2019 and 2026, I compare each of their 18.4 million duty statements with the duty statements of early 2019, and flag a statement as new when nothing in the early-2019 postings is close to it. A language model then classifies the flagged statements: whether they describe work on AI, whether that work is building AI or working with it, and whether they describe new tasks or familiar tasks with a new tool, object or wording.

Three main findings emerge on how jobs are changing. First, new work has grown where AI is used, and most of that growth is work on or with AI itself. In the third of occupations with the highest AI use according to the Anthropic Economic Index (Massenkoff and McCrory, 2026), duties that describe new tasks rose from under one in every 100 in 2022 to about three and a half in 2026, of which 72% are AI work (2.5 AI tasks against 1.0 non-AI tasks per 100 duty lines, with most of the non-AI rise occurring before ChatGPT); in the third of occupations with the least AI use, new tasks stayed below one in 200. Most newly worded duties outside AI are variations or re-wordings of familiar work rather than new tasks, and five of the six largest new or fastest-rising job titles are AI titles. Second, the mix of AI work is shifting from building AI towards working with it. Developing, deploying and selling AI systems still account for most AI duties, but their share has fallen since 2023: duties that ask workers to use AI tools, direct or review AI output, or lead AI adoption grew from 4% of AI duties in 2019–23 to 22% in 2026, and outside computer occupations they now make up a large share of AI work. Third, occupations are taking on technical and data tools that other occupations already used, such as production managers building dashboards (0.6% of their postings in 2022 to 4.7% in 2025–26) and financial analysts writing code to automate analysis: across all occupations, the share of postings naming a data, code or workflow-automation tool that the occupation rarely used in 2019–20 more than tripled between 2022 and 2026, rising most in occupations that use AI more.

Two further results concern how AI work is priced and how well official task inventories capture it. First, postings with AI duties advertise higher pay (0.17 log points more for building AI and 0.05 for working with AI within the same occupation, state and year), largely because higher-paying firms and dedicated AI titles account for most of them: within the same firm and occupation the premium is 0.07 log points for building AI and 0.02 for working with AI, and within the same job title at the same firm and year it is one or two per cent. Second, O*NET's occupational task lists—the standard task inventory on which most AI exposure measures are built—have not kept pace with the new work in job postings. Even in its August 2026 release, O*NET has no task statement about language models or working with AI, and for only about half of the new AI duties in job postings does the closest task in their occupation's O*NET list describe related work (against 89% of continuing duties), usually an older analogue such as programming robots or building business-intelligence tools. Pre-existing exposure scores built on O*NET still predict which occupations gain new work, because employers are adding AI tasks in the same occupations whose existing tasks are exposed to AI, but static task lists miss how work is changing within those jobs.

So far, the new tasks that generative AI has brought into job postings are mainly tasks that operate the technology itself, shifting from the activities of building it to those of working with it. Job

postings cannot yet show whether those tasks will last or are themselves a passing stage before further automation.

The paper also contributes data. Research on job postings relies mostly on commercial datasets that are costly and cannot be redistributed (Hershbein and Kahn, 2018; Deming and Kahn, 2018; Atalay et al., 2020; Acemoglu et al., 2022). The postings used here come from Common Crawl's public web archive, and I will release the build code together with the posting-level dataset: for each posting, its metadata, occupation, industry and seniority codes, weights, and posting-level measures of new work and AI duties, alongside aggregate series by occupation, state and quarter. Although these postings capture a selective share of US hiring tilted towards professional and technical jobs, they track official statistics and commercial postings across states and detailed occupations and in posted pay, follow the levels of remote-work and pay-transparency trends, and respond to shocks such as Colorado's pay-transparency law and the pandemic shift to remote work (Section 2 and Appendix A).

The closest work uses job postings to study generative AI. Chen et al. (2024) count, in Lightcast postings to 2024, the skills from Lightcast's taxonomy that are new to a firm and occupation, and find more of them in occupations where AI augments work; I classify the duty statements themselves and test whether the new lines describe new tasks (Appendix C compares the two counts). Koll et al. (2026) count skills in Lightcast's taxonomy that are new since 2010–11 and price them, including AI developer and user skills, through 2024; Chin et al. (2026) map ChatGPT messages to O*NET activities and find that many concern another occupation's tasks. Where skill taxonomies record mostly tools, credentials and subjects, duty statements describe what the worker will do, allowing this paper to test whether newly worded duties represent new tasks and to follow them through 2026. Wang et al. (2026) score the exposure of tasks extracted from Lightcast postings and decompose its decline into hiring shifts and changes within jobs, and Shukla (2026) maps 288 million postings line by line to O*NET tasks to measure the work AI cannot do; both measure the AI exposure of the tasks that postings list, whereas this paper measures the duties that are new. Hosseini and Lichtinger (2026) identify firms that adopt generative AI by their first posting for workers who integrate it, and find that junior employment falls at adopters; Chandar and Klein Teeselink (2026) find similar shifts across 41 countries. On new work, Marguerit (2025) measures it with new O*NET job titles before generative AI, and Humlum and Vestergaard (2025) find in Danish surveys that chatbots create some new tasks; Acemoglu et al. (2022) study earlier AI in vacancies. Studies of the job ladder ask whether AI reduces junior hiring (Brynjolfsson et al., 2025a; Hosseini and Lichtinger, 2026; Iscenko and Curto Millet, 2026). Meisenbacher et al. (2025) release aggregate O*NET features from National Labor Exchange postings; the data released here are at the level of the individual job posting.

Section 2 describes the data and Section 3 the measurement pipeline. Section 4 presents the results, Section 5 discusses the limitations, and Section 6 concludes.

2 Data

2.1 Postings from applicant tracking systems

Large employers publish their job adverts through applicant tracking systems: software that hosts the advert and collects applications, such as Greenhouse, iCIMS, Workday or Lever. An ATS can be detected for 97% of the Fortune 500 (Purcell, 2026). ATS job pages sit at predictable web

addresses, and Common Crawl, a public archive of the web, has captured many of them since 2017 (Common Crawl Foundation, 2026). I start from 20 widely used ATS platforms. Eighteen of them have enough archived job pages to test: for each, I fetched 20 random archived job pages in each of four years (2018, 2021, 2024 and 2026) and checked what could be extracted. I keep the 14 platforms whose pages reliably give at least a title and a location, and query the archive's index for their job-page addresses in all 96 crawls from January 2017 to September 2026. They hold 84–95% of the US job pages that the archive holds for the 20 platforms (Table A1).

Each page is parsed twice, once for structured data (a schema.org `JobPosting` object) and once with a template for the platform, and each field comes from the structured data where it exists. The 10.5 million captured pages collapse to 5.2 million jobs by platform, employer and job identifier. Employers often repost an advert under a new identifier, so jobs with the same title, location and text are merged; the merge removes 7.9% of US jobs. A posting is classified as US from its location or, failing that, from the page metadata, the text, its other captures or the employer's other postings. The result is 3,341,872 US postings from 52,473 employers (Table A2). Each posting has its title, employer, location, pay where stated, text, the dates it was posted, first seen and last seen, and a code in the 2018 Standard Occupational Classification. I date postings by their stated date of posting where it falls within the year before the crawl that first captured them, which holds for 88% of them, and by the first capture otherwise; the first capture comes a median of 37 to 67 days after the stated date, depending on the year, and more than a year after it for between one posting in twenty-five and one in eight (Appendix D). Dating every posting by its first capture instead changes no result (Appendix D).

Occupation codes come from a text classifier trained on 63,000 postings coded by a language model (Claude Opus 5.5) from the title, the employer and the posting's duty and role sentences, and on postings whose titles match O*NET titles. On held-out postings it gives the same detailed code as the language model for 74% of postings and the same major group for 87%, about as often as a second language model agrees with the first (73%), and postings with AI or new job titles are coded as well (78%). Every posting's detailed code places it in a third of occupations by AI use and gives it an O*NET task list; the occupation-level design uses only codes the classifier is confident about, which cover 64% of postings at 90% detailed agreement, a share that is flat over the years (64–68%) and ranges from 43% of production postings to 85% of health-care postings (Appendix A.3).

The postings capture a small, selective share of US hiring (0.5 postings in 2024 for every 100 hires in the Job Openings and Labor Turnover Survey). Even a complete census of online adverts would lie well below 100 postings per 100 hires, as employers often fill multiple vacancies from a single advert and many hires occur without an online posting. Beyond that, Common Crawl archives a sample of the web in each crawl, capturing only a fraction of each platform's job pages; these 14 platforms cover part of the ATS market (and Workday's pages carry text only from 2022); and many employers, notably school districts, public employers and health systems, post elsewhere. Like all online vacancies, these postings over-represent professional work: to match the occupation mix of employment, 34% of these postings would have to change major group, against 21% of Lightcast's commercial postings (see also Hershbein and Kahn, 2018).

Where the archive captures postings, their geographic, occupational and pay distributions match external benchmarks (Figure A1). State shares correlate with employment shares at 0.92–0.98 a year; across detailed occupations, log posting shares correlate with Lightcast's at 0.74, and median posted pay with the occupation's median wage at 0.89; titles agree with an independent parse of the same pages for 99.9% of postings. Over time, the shares of postings that state pay or offer remote work follow the trends of Indeed's and Lightcast's series (correlations in levels of 0.85–0.98), though

not their quarter-to-quarter changes, and the remote share runs above both from 2021, and further above from 2024. Both series respond to known shocks: when Colorado required pay ranges in 2021, the share of its postings with pay rose from 10% to 55%, and the share offering remote work rose from 3% to 13% between early 2020 and late 2021. Appendix A reports these benchmarks and the data's known gaps. As the sample over-represents computer and business occupations, I report results by occupation group and check the aggregates with representativeness weights.

2.2 Duty statements

The unit of measurement is a duty statement: a sentence in which the employer lists something the new hire will do. A text layer restores the line structure of descriptions published as flat text (every Workday description, and some others), types each section header, splits the text into sentences and classifies each sentence as a duty, a requirement, a statement about the role, the company, pay and conditions, or process and legal text (Appendix A). Duties are further classed as specific duties, broad responsibilities or generic statements ("other duties as assigned"), and generic ones are dropped. The classifier is trained on the 45,000 sentences of 1,200 postings, every one of which Claude Sonnet 5.5 labelled while reading the whole advert; in cross-validation that holds out whole postings, it finds Sonnet's duties with precision 0.92 and recall 0.94, and precision is at least 0.89 in every year (Table A8) and at least 0.88 on every platform. Throughout, a duty (or duty line) is the text, one duty sentence of one posting; a task is the activity it describes, which many duties in different postings and words can share; and new work means duties that describe new tasks. Duty statements are close to O*NET's task statements, though a duty can also name a tool, a setting or a standard. At the classifier's precision, about one line in twelve is not a duty—most often a general statement about the role—and the language-model reading of the new lines (Section 3.3) flags these false positives.

2.3 Analysis sample

I use postings dated from 2019 to the third quarter of 2026. The crawls of 2017–18 captured some platforms at several times their later rate, so this paper does not use them; the last such crawl, in January 2019, falls in the reference half-year, and 23% of the reference's distinct duty sentences appear only in it; and Appendix B shows the measure without it. I keep postings in English with a description of at least 200 characters, drop bulk posters (employer-years with at least 1,000 postings under at most 50 titles; scaling the cutoff to the nine months of 2026 removes no further employer), keep one posting per multi-location advert (the same normalised title and text at several locations, keeping the earliest), and keep postings with at least one duty line. Appendix Table A4 counts the postings each step removes. The sample retains 28,835 postings whose state could not be resolved (dropping them changes no result; Appendix D) and 17,338 postings by recruitment agencies, which list real duties. The resulting sample has 1,629,317 postings from 48,304 employer accounts and 18.4 million duty lines, of which 10.1 million are distinct sentences (Table 1).

Three further fields come from language-model labels extended by classifiers. Seniority comes from the job title: a model trained on 30,000 titles labelled by Sonnet 5.5 assigns one of eight levels from intern to executive, agreeing with the labels on 88% of held-out titles. Industry is a two-digit NAICS sector for each firm with at least three postings, coded by Sonnet 5.5 from the firm's name, titles and occupation mix, and recoded from its company-description sentences where Sonnet was not confident; it covers 99% of postings. Firms that moved between platforms are linked by name,

a normalised name unique to one account on each of at least two platforms, which joins 48,304 employer accounts into 45,678 firms: 5,040 accounts become 2,414 firms, and the firm fixed effects of Section 4 use this identifier.

Table 1: Analysis sample by year

Year	Postings	Employers	Duty lines (M)	Lines per posting	New (%)	New AI (%)
2019	186,588	9,838	1.97	10.5	10.0[b]	0.11[b]
2020	133,435	9,577	1.39	10.4	11.8	0.15
2021	225,174	12,485	2.44	10.8	12.9	0.19
2022	179,575	13,227	2.00	11.1	14.2	0.15
2023	133,581	12,579	1.52	11.4	14.9	0.21
2024	263,295	17,474	3.02	11.5	14.4	0.36
2025	287,258	18,521	3.32	11.6	15.0	0.90
2026[a]	220,411	14,648	2.72	12.3	17.4	1.69
All	1,629,317		18.37	11.3		

Notes: Postings in English with a description of at least 200 characters and at least one duty line, excluding bulk posters, one posting per multi-location advert. Employers: employer accounts (platform and employer identifier). New: duty lines with no close counterpart among the duty sentences of 2019H1 postings (Section 3.1). New AI: new duty lines that describe AI work. [a]January to September. [b]2019H2 only: no line of the reference half-year is new.

2.4 Release

With a later version of this paper, I will release a posting-level dataset covering all 3,341,872 US postings, together with aggregate series by quarter, occupation, state and platform, and the code for the build. For each posting, the release will provide its dates, location, pay fields, occupation code with its confidence, seniority, industry, crawl-intensity and representativeness weights, flags for agencies, bulk posters, remote work and closed adverts, and posting-level summary measures of duty lines, novelty, AI work by side and tool use. The raw text of the adverts, which is the employers' own writing, will not be redistributed; the data will be released under CC BY 4.0 and the code under the MIT licence.

3 Measuring new work

Measuring new work in 18.4 million duty lines takes four steps (Table 2). A screen first flags the duty lines that have no close counterpart among the duties of early 2019; it compares sentence embeddings, so it can run on every line, and it is deliberately loose. Language models then do three jobs: they label every duty sentence that mentions AI with the side of AI work it is on; they read random samples of the flagged lines, each beside its closest duty of 2019, and judge which describe new tasks; and they grade how closely duty lines match O*NET's task statements. Classifiers trained on the models' labels extend the AI labels to every line, and a second model re-reads a random part of most labelling tasks (Appendix B.6).

Table 2: How new work is measured

Step	Question	Method	Applied to
1. Screen (Section 3.1)	Does the duty line have a close counterpart among the duty sentences of 2019H1?	Sentence embeddings; a similarity threshold set on 2019H2	Every duty line
2. AI labels (Section 3.2)	Is the line about AI, and is it building AI or working with it?	Keyword rules; Sonnet 5.5 reads every keyword sentence; a classifier for the rest	Every duty line
3. Reading (Section 3.3)	Does a new line describe a new task, a familiar task with a new tool, object or setting, or a familiar task in new words, or is it not a duty?	Sonnet 5.5, beside the closest duty of 2019H1	Random samples of new lines
4. O*NET (Section 3.4)	Does the occupation's O*NET list describe related work?	Sentence embeddings; Sonnet 5.5 grades samples of matches	Every duty line

3.1 Duty lines with no counterpart in 2019

A duty line is new if nothing in the postings of early 2019 is close to it. I embed every distinct duty sentence with a small sentence-embedding model (all-MiniLM-L6-v2; Reimers and Gurevych, 2019; Wang et al., 2020), which places sentences of similar meaning close together, and compute for each its highest cosine similarity to any of the 679,314 distinct duty sentences in postings of the first half of 2019. The first half of 2019 is the earliest stretch of usable postings and comes before both COVID-19 and generative AI; a reference of all of 2019 would make the pandemic year the first to be compared with it.

The threshold is set by what the screen finds when there is little new work to find. Even then, some duty lines have no close counterpart in the reference, because employers word the same duties in many ways and the reference holds only a sample of postings. The second half of 2019, six months after the reference, shows how many: the threshold, 0.66, is the similarity below which 10% of its duty lines fall. This baseline churn assumes that 2019H2 holds little new work, which the reading confirms: Sonnet reads 1.6% of its new non-AI lines as new tasks (Section 4.3). The screen is deliberately loose, letting through a tenth of the lines of a period with almost no new work so as to miss as few new tasks as it can; the reading then decides which of the lines it lets through are new work (Section 3.3). Sonnet's reading of pairs across the range of similarity shows what the threshold means (Appendix Figure B1): most pairs below it describe familiar tasks with a new tool, object or setting, new tasks above it are rare, and the screen passes most of the non-AI lines near the threshold that Sonnet would call new tasks, though fewer of those written in familiar words (Section 4.3 and Appendix B). Thresholds that make 5 to 20% of 2019H2 new move the levels of new work, read or not, but not the comparisons between occupations (Appendix Table C1 and Appendix D).

The share of new lines rises with distance from the reference, to about 14% by 2022, and it rises as much in the occupations that use AI least as in those that use it most. Some of this is genuine change, but most is variation in the objects, tools and wording of familiar tasks: of the rise in new non-AI lines between 2019H2 and 2022 (4.2 points), Sonnet reads 6% as new tasks and the rest as variants, re-wordings and lines that are not duties (Section 4.3). Measured against the duties of the two previous years instead, in windows of the reference's size, new non-AI lines do

not rise, while new AI lines still do (Appendix B). Given this wording drift, I measure new work from the language-model reading and compare occupations with one another, rather than treating the raw share of unmatched lines as new work; the comparisons also hold in the firms that posted throughout (Appendix C).

Duties from the COVID-19 pandemic provide a check on whether the pipeline detects a known wave of new tasks. Of the duty lines of 2020–21 that mention COVID-19, 67.7% are new, and Sonnet reads 38% of these as new tasks, against a few per cent of other new non-AI lines; clustered into themes, the new lines of 2020–21 recover COVID-19 duties, from screening and safety protocols to vaccination programmes (Appendix B).

3.2 AI work: building it or working with it

I find duty lines about AI in two steps. Keyword rules flag every duty sentence that mentions AI, machine learning, language models, generative AI or named AI tools. Claude Sonnet 5.5 then reads every one of the 134,656 flagged sentences, with the job title and occupation group but not the date, and records whether it describes AI work, which side of the technology it is on and what the worker does. Building AI covers developing, training and evaluating models and AI products; deploying and integrating them; and selling and supporting them. Working with AI covers using AI tools in one's own work; prompting, directing or reviewing AI systems and their output; and finding uses for AI, redesigning workflows around it and training colleagues in it. Governing AI (policy, risk and compliance) is a third side. These sides classify the worker's activity rather than the employer: a bank's data scientist training a fraud model is building AI, whereas an AI vendor's customer-success staff helping client firms adopt its software are working with AI (a sentence that mentions both takes the side it is mainly about).

A second model (Claude Opus 5.5) independently reads 2,000 of these sentences and agrees with Sonnet on the side of AI work in 90% of cases. Duties about building AI often name specific methods rather than "AI" itself, so the keyword rules catch about half of building-AI work compared with most working-with-AI duties from 2024 on. I account for this in two ways (Appendix B.3): a classifier trained on all labelled sentences extends the AI labels to every duty line in the corpus, and the overall split between building AI and working with it is reported after dividing each side's keyword lines by its keyword catch rate, estimated on random samples of keyword-free lines.

3.3 Reading the new lines

Every label in this paper comes from a census or a designed random sample (Appendix Table B7). Sonnet 5.5 reads random samples of the new lines, whether about AI or not, each paired with its closest duty from early 2019 and stripped of its year and occupation cell, and judges whether it describes a new task, a familiar task with a new tool, object or setting, or a familiar task in new words, or is not a duty at all. The sample covers 500 new non-AI lines in every year from 2019H2 to 2026 and every third of occupations by AI use (plus 250 a year in occupations without a measure of AI use), alongside 400 new AI lines on each side of AI work in 2023–24, 2025 and 2026. The lines are read in batches of 40 that mix AI and non-AI lines from every year and third; read only beside other AI lines, AI lines are judged new tasks less often (Appendix B.6). A further reading of 400 lines per side from the third that uses AI least, and of AI lines of 2019–22, checks the shares that the main estimate pools across thirds or borrows from 2023–24 (Appendix D). A new line counts as new work only when it is read as a new task, whether it is about AI or not. A second

model, Opus 5.5, reads a fifth of the same batches; it agrees with most of the readings and calls fewer non-AI lines new tasks (Appendix B.6). The themes of the new lines give a second reading: I cluster the 416,000 distinct new sentences of 2025–26 into 1,500 themes and keep those whose share of employers in 2025–26 is at least three times that of 2019–22. Sonnet names 90 of them, the 60 that rose most and the largest whose lines are mostly not about AI, and judges each to be AI work by side or, if not, a new task, a variant, a re-wording or not work; Opus re-reads them and judges the AI themes on the same four kinds.

I group occupations by observed AI use, the observed-exposure measure of Massenkoff and McCrory (2026): it is the share of the occupation's task time in tasks that are rated feasible for language models by Eloundou et al. (2024) and that see work-related use in conversations with Claude in 2025. It is therefore measured after the period of interest began, and it shares its task ratings with Eloundou et al.'s exposure measure; Appendix D repeats the main comparisons on thirds of that measure, which is rated before the period. Thirds are weighted by postings, so each holds a third of them: the third that uses AI most is 72 occupations, the third that uses it least 459.

3.4 Duty lines and O*NET tasks

O*NET's task statements are the standard list of what each occupation does, and most measures of AI exposure are built on them, so those measures can see new work only if the lists describe it. To ask whether they do, I map every duty sentence to its closest O*NET 27.1 task statement (2022) with the same embeddings, both among all 19,265 statements and among those of the posting's own occupation. A duty line has a related O*NET task when the closest task is similar enough that a language model is more likely than not to judge the two to describe at least related work: Sonnet 5.5 graded 1,530 such pairs spread over the range of similarity, half from each lookup, and each lookup has its own threshold, where a logistic fit of its judgements crosses one half (Appendix Figure B8). The most similar task is not always the most related one, so Sonnet also grades the five closest tasks of random samples of lines on both sides of the threshold, and, as a benchmark, five tasks drawn at random from the same list. The mapping gives an outside reference that does not move with the sample: a duty with no related O*NET task is new relative to the official description of work in 2022, not relative to early-2019 postings.

4 Results

4.1 How much new work

New work has grown where AI is used (Figure 1). Most new duty lines are not new work. Even in 2019H2, when there is little new work to find, a tenth of duty lines have no close counterpart in early-2019 postings, and Sonnet reads almost all of them as variants, re-wordings or lines that are not duties. This churn grows with time since the reference and still makes up most of the new lines in 2026. Genuinely new work, the new lines that Sonnet reads as new tasks, is a small part of them: across all occupations it rose from 0.4 in every 100 duty lines in 2022 to 1.7 in 2026 (panel a). In the third of occupations that use AI most it rose from 0.7 to 3.5, 72% of it AI work, while in the third that uses AI least it stayed below one in 200 (panels b and c). Not every new AI line counts: many vary AI work that postings already described in 2019, such as building language-model pipelines rather than machine-learning ones, and only the lines read as new tasks count as new work. Most

of those are familiar activities done with or on AI, such as drafting or coding with an assistant, rather than new activities such as directing agents (Appendix B.6). Appendix D gives the intervals around these figures and the readings behind them.

A map of duty space shows the same pattern: between 2019–22 and 2025–26, holding the occupation mix fixed, one cluster of AI work gained more share of duty lines than any other part of the map (Appendix B.4). Computer occupations hold 74% of the new AI lines of the high-use third in 2026, but the contrast between the thirds holds without them and under other weights (Appendix Table C1).

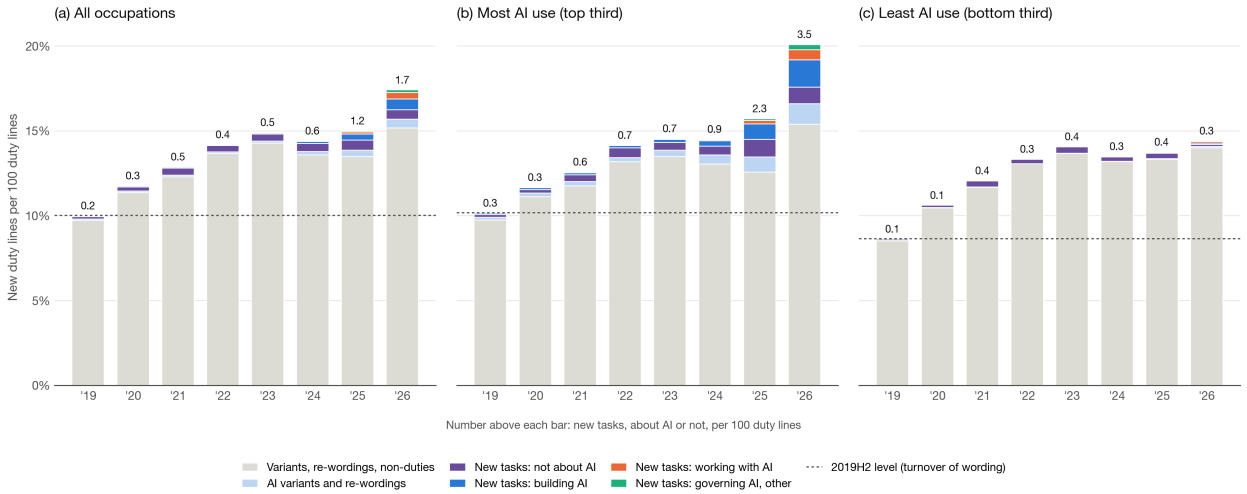


Figure 1: New work in duty statements. New duty lines (no close counterpart among the duty sentences of 2019H1 postings) per 100 duty lines, by year (2019: second half; 2026: January to September), split by what Sonnet 5.5 reads them to be: variants and re-wordings of familiar tasks and lines that are not duties, not about AI (grey); AI lines that vary or re-word AI work already described in 2019 (pale blue); new tasks that are not about AI; and new AI tasks by side. The split uses Sonnet's reading of 500 new non-AI lines in every year and third of occupations by AI use and of 400 new AI lines on each side in 2023–24, 2025 and 2026, read in batches that mix the samples (2019–22: the shares of 2023–24; governing and other AI lines: the pooled share of the two sides; Appendix D gives the series with those shares measured directly, and 95% intervals). (a) All occupations. (b, c) The thirds of occupations with the most and least observed AI use. Dotted line: the panel's 2019H2 level of new lines, the turnover of wording (10% in panel a, by construction). Numbers above the bars: new tasks, about AI or not, per 100 duty lines.

4.2 Building AI or working with it

The mix of AI work is shifting from building AI towards working with it, although both sides are growing. The sides classify activities rather than employers (Section 3.2), following the distinction between AI developer and AI user skills in Maslej et al. (2025) and Koll et al. (2026) but measured in duties rather than skill tags. Figure 2 shows the share of postings with at least one duty of each side, by occupation group. Building AI grew first and most in computer occupations: from about one in ten of their postings in 2019–22 to 34.7% in the second quarter of 2026. Working with AI barely existed before 2024. By mid-2026 it appears in 13.9% of postings in computer occupations, 11.2% in management and business occupations and 2.1% in the rest; outside computer occupations it is now a large part of AI work.

Figure 3 shows every major occupation group with at least 1,000 postings in both years, on a scale

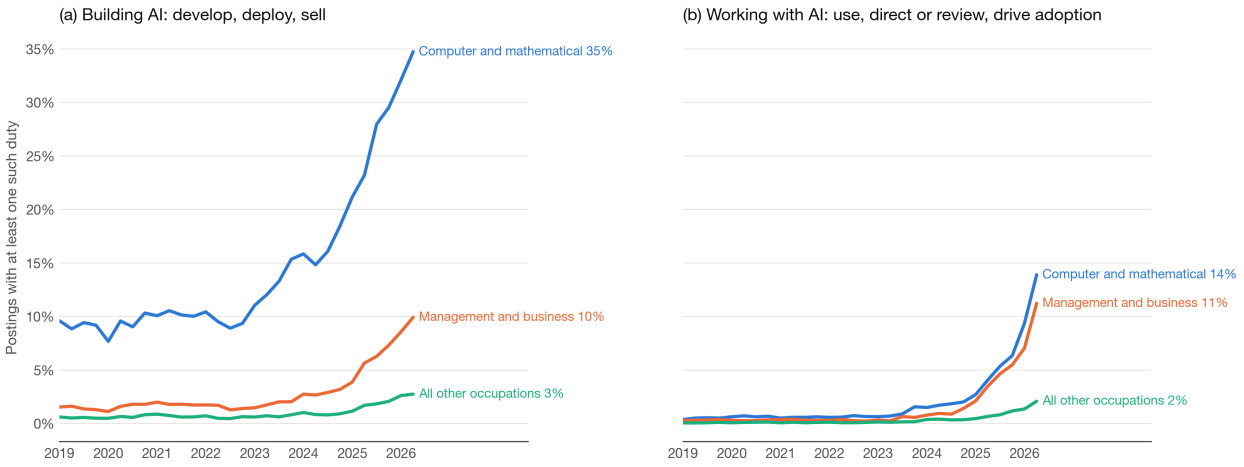


Figure 2: Share of postings with at least one duty line about building AI (a) or working with AI (b), by quarter and occupation group. Every duty line is classified: Sonnet 5.5 labels for AI-keyword sentences and a classifier trained on all labelled sentences for the rest. 2026Q3, a partial quarter, is omitted.

that does not depend on the group's size. AI duty lines rose in every professional and technical group, most in computer occupations (12.9 per 100 duty lines in 2026). Legal occupations, which had almost no AI duties in 2022, come second, just ahead of management (3.8): they rose from 0.18 to 3.96 AI lines per 100 duty lines, with a working-with-AI duty in 6.5% of their 2026 postings, while titles such as legal engineer and legal operations manager grew from about 18 postings a year in 2022–23 to 85 in 2026. The groups with little AI work are those whose work is done by hand or in person: health-care support, production, transportation, construction and food preparation.

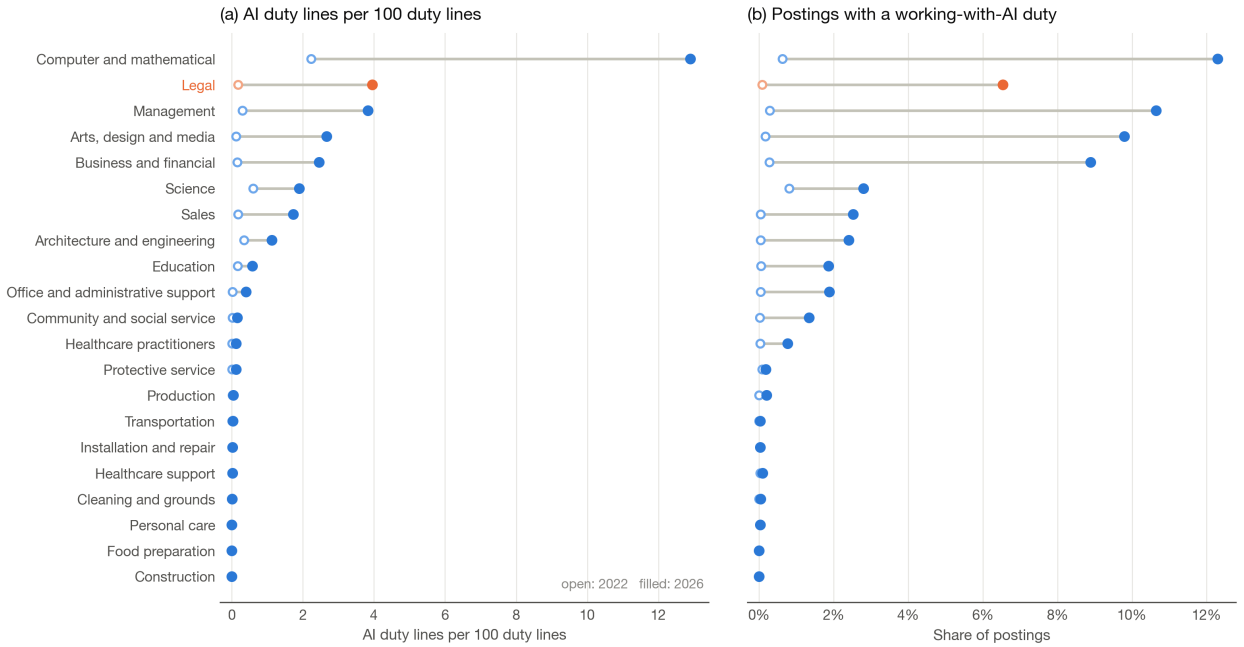


Figure 3: AI duties by occupation group, 2022 (open) and 2026 (filled). (a) AI duty lines per 100 duty lines, every duty line counted (Sonnet 5.5 labels for keyword sentences, a classifier for the rest). (b) Share of postings with a working-with-AI duty. SOC major groups with at least 1,000 postings in both years, sorted by panel (a) in 2026; legal occupations in orange.

The specific activities within AI duties have shifted alongside this broader mix (Figure 4). AI work grew from 0.4 in every 100 duty lines in 2019 to 3.3 in 2026 (panel a). Among AI duty lines with a keyword, building models and AI products made up 73% in 2019 and 33% in 2026, a smaller share of a much larger total. Growth came in driving adoption, from 2% to 17%; using AI tools, from 3% to 14%; selling and supporting AI, from 6% to 12%; and governing AI, from under 1% to 7% (panel b). Allowing for the share of each side's lines that the keywords catch (Appendix B.3), AI work was 4.2 in every 100 duty lines in 2026, and working with AI rose from 4% of it in 2019–23, when the share was flat and few keyword-free lines described such work, to 22% in 2026 (95% interval 20.7–22.8%); building AI is still the larger side, at 2.7 lines in every 100 against 0.9. Surveys find generative AI adopted at work faster than earlier technologies (Bick et al., 2024), and experiments find that it raises the productivity of the people who use it (Brynjolfsson et al., 2025b; Noy and Zhang, 2023); the duties that now ask for it are the demand side of that adoption.

The balance differs by sector (every duty line counted, with the classifier's labels; on AI lines per 100 duty lines, the catch-rate correction raises the building side everywhere and changes no ordering, Appendix D). In 2026, 27.1% of postings in information firms have a building-AI duty and 15.6% a working-with-AI duty, and in manufacturing 7.2% against 3.6%; in finance, health care and retail the two are much closer (8.9% against 9.1%; 1.7% against 2.0%; 1.8% against 2.3%).

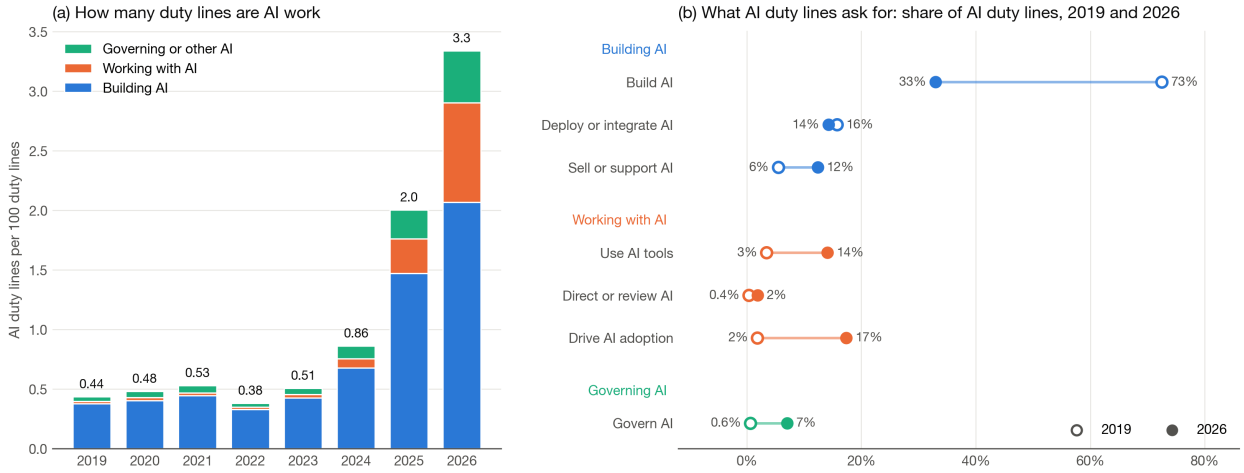


Figure 4: What AI duties ask for. (a) AI duty lines per 100 duty lines, by year and side; every duty line is classified (Sonnet 5.5 labels of sentences with an AI keyword, a classifier for the rest). (b) The activities of AI duty lines as shares of all AI duty lines, 2019 (open) and 2026 (filled): Sonnet 5.5 labels of every duty sentence with an AI keyword; the shares exclude AI work that fits no activity and are not corrected for the keywords' catch rates (Appendix B.3). Appendix Table C2 gives every year.

4.3 New work outside AI

Most of the new lines are not about AI, and most of those are not new work (Figure 5). In every year, Sonnet reads 67–71% of the new non-AI lines as familiar tasks with a new tool, object or setting, 13–15% as familiar tasks in new words and 13–15% as not duties at all. New tasks are 1.6% of the new non-AI lines of 2019H2, when little new work is to be expected, and 3.5% of those of 2026. In the occupations that use AI most they rose from 1.7% in 2019H2 to 4.3% in 2022, peaked at 7.5% in 2025 and were 5.9% in 2026, so most of the rise came before ChatGPT; in those that use it least they stayed at a few per cent. As a share of all duty lines in high-use occupations, new non-AI tasks were 0.6% in 2022 and 1.0% in 2026, against 2.5% for new AI tasks in 2026. These

counts likely understate new non-AI work, which is more often written in familiar words. In areas of work known to have emerged since 2019, such as climate disclosure and new compliance regimes, the embedding screen passes only 48–69% of the lines that Sonnet would call new tasks (Appendix Table B1), and the second reader calls fewer lines new tasks. Scaling by those recall rates across the two readers places new non-AI tasks in high-use occupations in 2026 between 0.8 and 2.8 per 100 duty lines, against 2.5–2.9 for new AI tasks—of the same order in levels, but rising by 0.3–1.1 points since 2022 against 2.4–2.7 for AI tasks, so most of the recent growth of new work where AI is used is AI work (Appendix D).

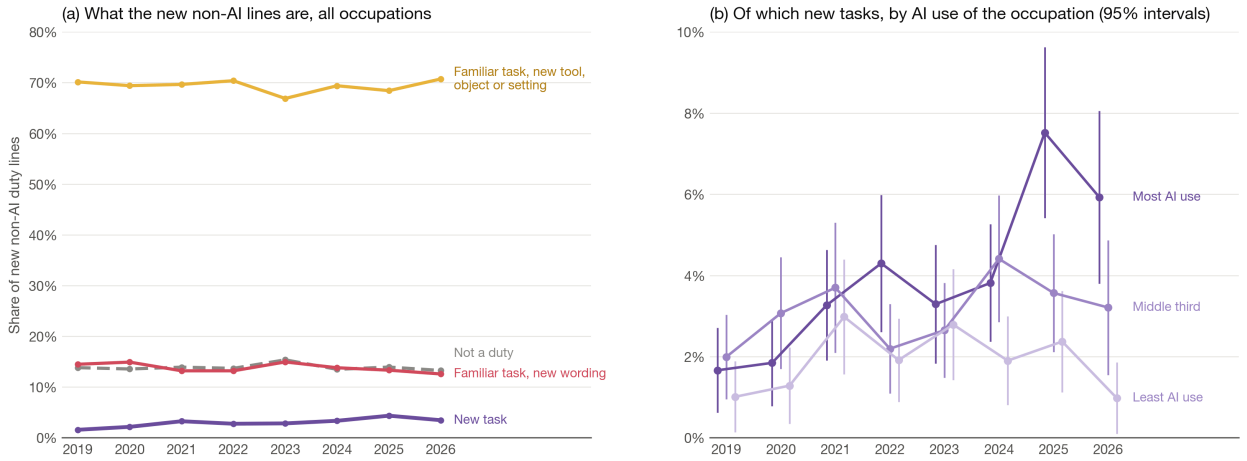


Figure 5: What the new non-AI duty lines are. Sonnet 5.5 compares each of 500 new non-AI lines in every year and third of occupations by AI use with the most similar duty sentence of 2019H1, without seeing the year or the cell (Appendix B.6). (a) Share of each reading, all occupations: a new task; a familiar task with a new tool, object or setting (such as scanning inventory with a smartphone app, beside 2019's handheld scanner); a familiar task in new words (such as taking care of permits, beside securing them); or not a duty (examples paraphrased). The thirds and the occupations without a measure of AI use are weighted by their new non-AI lines. (b) The share read as new tasks, the lowest line of panel (a), by third of observed AI use, with 95% intervals. 2019: second half.

Clustering the new duty lines into themes tells the same story (Figure 6). Of the 60 themes that rose most, 49 are AI work: integrating language models into applications, building AI platforms and agents, using AI tools to work faster, building AI-native workflows and championing AI adoption (panel a). Of the 40 named themes that are not AI work, Sonnet judges all but one—building automated security and compliance guardrails—to be familiar work in new words or templates, variants of familiar work, or not work at all (Appendix Table C6). Judged by Opus on the same four categories, 64% of the AI themes are new tasks (93% of the working-with-AI themes and 52% of the building themes; Appendix B.6), and running the same clustering on 2020–21 recovers COVID-19 duties. The new non-AI tasks are too scattered to form rising themes, so panel (b) sorts those of 2023–26 by kind. The largest group is technology outside AI (43%), such as blockchain and no-code tools, followed by new products and markets (26%), such as crypto assets and electric vehicles, and new rules and compliance regimes; some are AI work that the AI labels missed, such as compressing models or building systems that evaluate their output.

New job titles carry the most distinctive duties (Figure 7). Five of the six largest titles that are new since 2022 or have grown fastest are AI titles, and they rose steeply from late 2024: forward deployed engineers were posted by 9 employers in 2019–22 and 247 in 2024–26, AI engineers by 10 and 168, and GTM engineers, a title absent from these postings before 2024, by 118 (Appendix

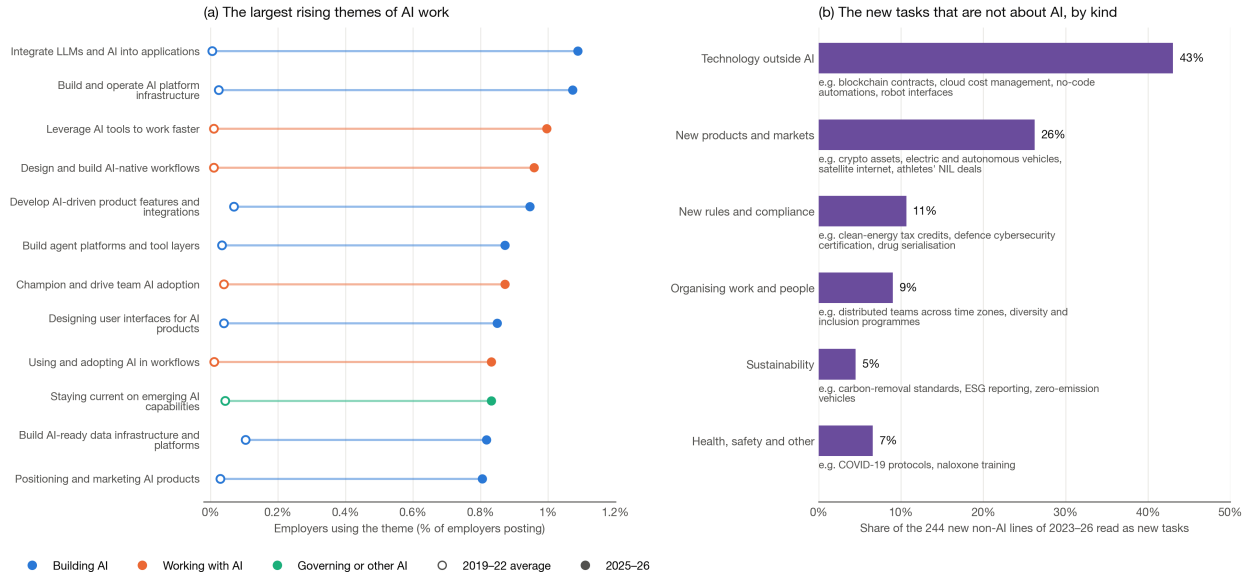


Figure 6: What the new work is. (a) The largest rising themes of AI work among the new duty lines: the share of employers using each theme in 2019–22 (open) and 2025–26 (filled). Themes: clusters of the distinct new sentences of 2025–26; rising: used by at least three times the share of employers. Names and sides by Sonnet 5.5 from recent and older example sentences; the themes not about AI are in Appendix Table C6. (b) The new non-AI lines of 2023–26 that Sonnet reads as new tasks (Figure 5), by Sonnet's judgement of what makes them new; examples paraphrased from the lines.

Table C7). By the first half of 2026, 9.1 in every 1,000 employers posting were hiring a forward deployed engineer. The new titles also come with duties unlike their neighbours': controlling for the size of the comparison set, the share of a new title's duty lines with no close counterpart among the duties of its three nearest established titles is 25 points higher than for an established title, a measure of distinctiveness from similar titles rather than of new work in the sense of Figure 1. New titles are rare: titles that at least three employers used in a year and that no posting of 2019–22 used, in any spelling, are 0.7% of 2026 postings (Appendix C).

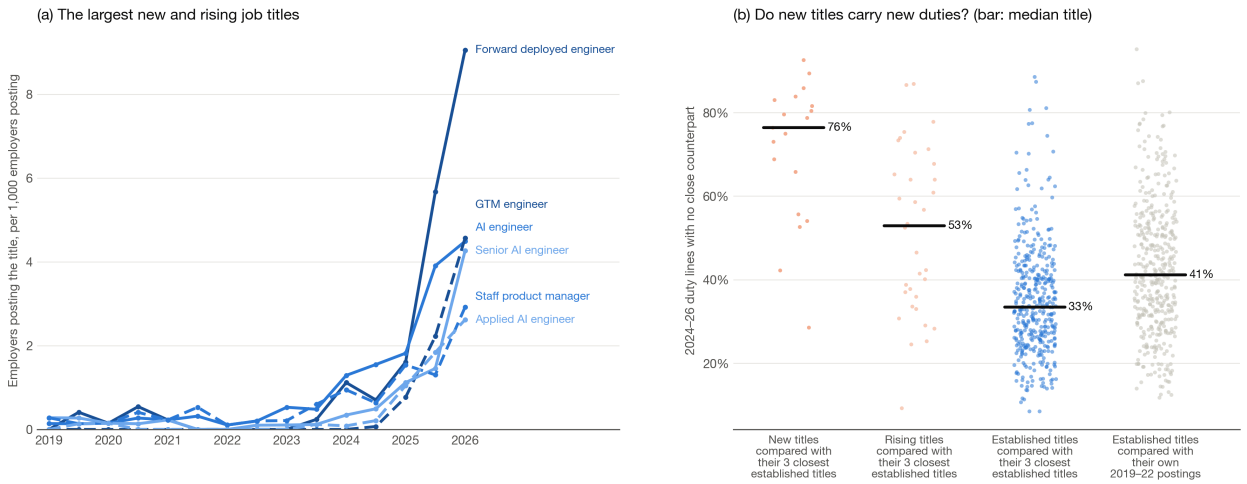


Figure 7: New job titles. (a) Employers posting the largest new and rising titles, per 1,000 employers posting in the half-year, 2019H1–2026H1 (Appendix Table C7 gives the counts). New titles: at least 10 employers in 2024–26 and none in 2019–22, in any spelling (ignoring word order, seniority words and initialisms) and in any posting of those years; rising titles: at least 20 employers in 2024–26 and five times their 2019–22 count. All but staff product manager are AI titles, with an AI duty in at least half of their postings. (b) Share of 2024–26 duty lines with no close counterpart among the duties of the three most similar established titles (new and rising titles, and established titles against each other) or of the title's own 2019–22 postings; bar: median title.

4.4 Tools new to the job

The new tasks of the previous sections are new to the economy: no duty of early 2019, in any occupation, describes them. A task can also be new to a job without being new to the economy, when an occupation takes on a task that other occupations already did, such as a production manager building the dashboards that analysts built before. That is one form augmentation may take, as tools let workers do tasks that used to need specialists. I measure one well-defined kind of it: tasks done with technical and data tools. Five families of tools are defined in advance: dashboards and business intelligence, databases, programming and statistical software, which follow O*NET's software categories of November 2022 and are defined by the tools that O*NET marks as most requested in postings and by current tools it does not yet list, and workflow automation, which O*NET lacks and which is defined by its tools. An occupation did not use a family before generative AI when fewer than 1% of its 2019–20 postings named one of its tools in a duty, and I count in every posting since whether a duty names such a tool. Opus 5.5 reads samples of the matched lines to measure how often the tool is part of the work, and the counts are corrected for it (Appendix B.2).

Postings that ask for tools their occupation did not use rose from 0.36% in 2022 to 1.24% in 2026, most of it after 2024 and mostly in dashboards and workflow automation (Figure 8a). The rise is larger where AI is used most (0.45% to 1.78%) than where it is used least (0.08% to 0.36%), though from its lower base the low-use third grew about as fast in relative terms. In 2025–26 these postings also ask the worker to use AI more often than other postings in the same occupations (16% against 4%, a gap of 10 points within the same firm and occupation). Production managers build dashboards to track quality and production (0.6% of their postings in 2022, 4.7% in 2025–26), financial analysts write Python, SQL and VBA to automate their analysis (0.1% to 1.0%) and management analysts build low-code workflows (panel b). The occupations that already used these tools ask for them more too, from 10.7% to 17.0% of their postings, so in levels most of the

spread of these tools is deeper use where they were already common. Business-intelligence tools had begun reaching managerial postings before generative AI (Koll et al., 2026), and on the side of use, ChatGPT users often ask for help with the tasks of other occupations (Chin et al., 2026). Appendix B.2 and Appendix D (Figure D1) show that this increase is not driven by mechanical drift from the 2019–20 baseline, platform composition or alternative cutoffs, and that dashboards and workflow automation grew several times faster between 2022 and 2026 than the other eleven O*NET software families.

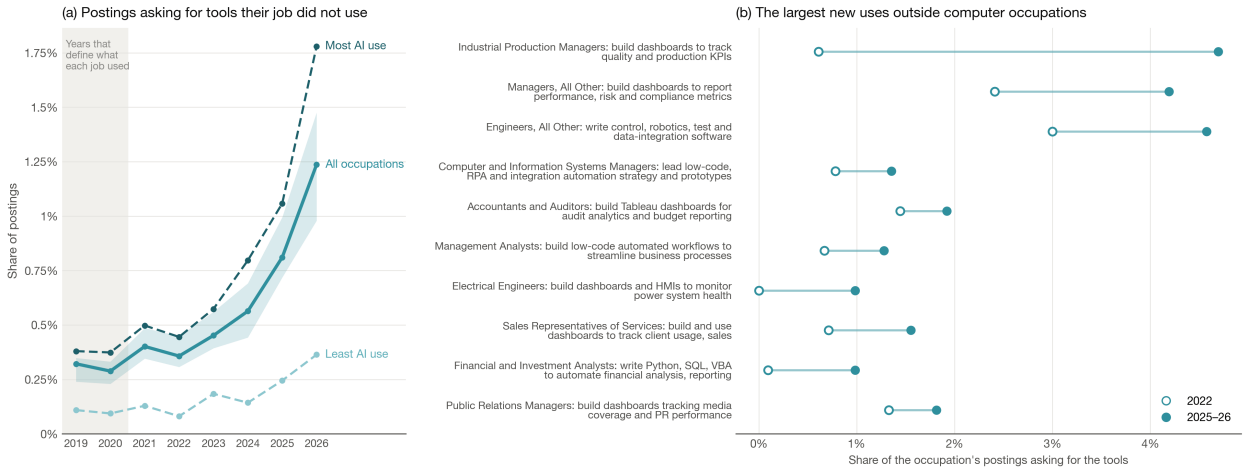


Figure 8: Technical and data tools new to the job. (a) Share of postings with a duty that names a tool from a family that the occupation did not use in 2019–20 (fewer than 1% of its postings), each weighted by the share of such matches that Opus 5.5 confirms (Appendix B.2): all occupations and the thirds with the most and least observed AI use; 145 detailed occupations with confident codes, weighted by their 2019–21 postings. Shaded: 95% interval from resampling employers and the validation sample. Grey: the years that define each occupation's earlier use. (b) The largest such occupation and tool pairs outside computer occupations whose recent lines Opus confirms, with Opus's summary of what the workers do: share of the occupation's postings in 2022 (open) and 2025–26 (filled).

4.5 New work and the official task lists

Most research on the labour-market effects of AI relies on O*NET's occupational task lists to define what each occupation does and which tasks AI can perform (Webb, 2020; Felten et al., 2021; Eloundou et al., 2024). Comparing the duties in job postings with O*NET's task statements shows that these official inventories lag behind the new work employers advertise. Of the 19,265 task statements in O*NET 27.1 (2022), only a handful mention artificial intelligence or machine learning, mostly as one technology in a list in engineering and science occupations, and none concerns language models or working with AI; O*NET 31.0, released in August 2026, nearly four years after ChatGPT, has 103 task statements that 27.1 lacked, none of them about AI.

When the duty statements of 2019–26 are matched to the O*NET 27.1 task list of their own occupation, 89% of continuing duties have a closest O*NET task that describes related work, compared with only 49% of new building-AI duties, 58% of new working-with-AI duties and 67% of new non-AI duties (Appendix Figure B7). Across all occupations, the O*NET tasks that the new AI duties are most often matched to are older analogues of the same broad activity, such as programming robots, creating business-intelligence tools or applying data-mining and machine-learning algorithms (Appendix Table B6). Appendix B.5 reports the matching and the grading

under stricter standards.

Static task lists therefore miss the new AI tasks emerging within jobs, even as exposure measures built on them predict which occupations are affected. Across detailed occupations, the share of duties that describe new tasks has a rank correlation of 0.69 with the pre-existing O*NET exposure score of Eloundou et al. (2024), because employers are adding AI work in the same occupations whose existing tasks are exposed to language models. Pre-existing O*NET measures thus identify the occupations where new work is appearing, but tracking how tasks change inside those occupations requires reading job postings directly.

4.6 Where new work lands

Do occupations where AI is used more acquire more new duties after generative AI arrives? I estimate, for 230 detailed occupations over half-years from 2019 to 2026,

$$y_{ot} = \alpha_o + \gamma_t + \sum_{s \neq 2022\text{H2}} \beta_s \, \mathbf{1}[t = s] \, U_o + \varepsilon_{ot},$$

where U_o is the occupation's observed AI use, standardised, and y_{ot} an outcome in percentage points. Occupations are weighted by their 2019–22 postings and standard errors clustered by occupation. Figure 9 shows the coefficients. Working-with-AI duties are flat in 2019–22 and rise from late 2024: by the second half of 2026 (to September), an occupation one standard deviation higher in AI use has 4.2 points more postings with such a duty. Averaged over 2024–26 against 2019–22 (2023 is in neither period) the effect is 1.7 points (standard error 0.4); it is about half that with major-group-by-period effects, 0.8 (0.4), or with occupations weighted by their employment rather than their postings, 0.79 (0.32) with OEWS employment and 0.60 (0.26) with new job starts in the CPS, and without the partial last half-year it is 1.1. New building-AI lines rise from 2023 and new non-AI lines only in 2026.

This cross-occupation regression is descriptive rather than causal. Occupations with higher observed AI use already differed before 2022 in their senior share, posting volumes and non-AI wording churn (panel c and Appendix Figure C2), and observed AI use correlates closely both with pre-existing task-exposure ratings (correlation 0.64; Eloundou et al., 2024) and with an occupation's 2019–21 share of remote postings, which absorbs the post-2022 gradient when entered alongside AI use (Appendix D).

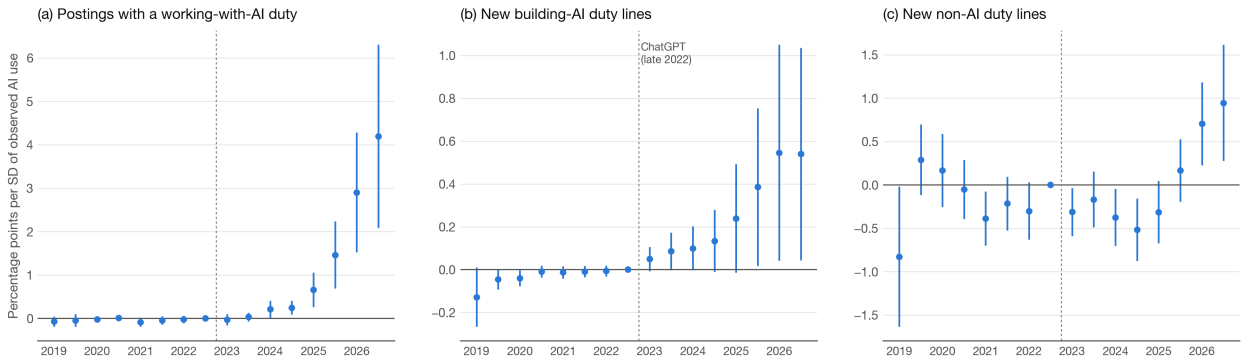


Figure 9: AI duties and new duty lines by observed AI use of the occupation: coefficients β_s on standardised AI use (Anthropic Economic Index) by half-year, relative to 2022H2, with 95% intervals. 230 occupations with confident codes, 3,570 occupation-half-years. (a) Postings with a working-with-AI duty. (b) New building-AI duty lines. (c) New non-AI duty lines, before reading.

4.7 Pay and seniority

Postings with AI duties advertise higher pay, and most of the premium is accounted for by which firms and job titles post them. Figure 10 decomposes the posted-pay premium of postings with a building-AI or a working-with-AI duty, on 428,836 postings that state pay, across successively tighter comparisons. Against postings in the same occupation, state and year, building-AI postings advertise 0.17 log points more and working-with-AI postings 0.05. Within the same firm, occupation and year the premia are 0.07 and 0.02; adding seniority and state to that cell gives 0.045 (standard error 0.004) and 0.007 (0.010). Within the same job title at the same firm in the same year they are 0.01 and 0.02, neither distinguishable from zero, but this comparison rests on the 1,751 and 880 postings in title cells where the indicator varies, and its interval for working with AI allows a premium of up to 5 per cent. Firms often introduce AI work through dedicated, high-paying job titles (Appendix Table C7), so conditioning on the exact title absorbs that channel rather than ruling out a pay premium. The estimates are much the same with the AI indicators taken from keyword lines alone, in states and dates where a law requires pay ranges, at either end of the posted range, in the firms that posted throughout, and larger for building AI after correcting for the recall of the AI labels (Appendix D); they line up with the premia for AI developer and AI user skills in Lightcast postings (Koll et al., 2026). Appendix Table C4 gives the occupation groups.

The postings are also becoming more senior: the share of junior titles fell from 24% of postings in 2022 to 20% in 2026, and the share of senior and staff titles rose from 13% to 19%. The shift began in 2024. It was larger in occupations that use AI more, but those occupations were already becoming more senior before 2022 (Appendix Figure C2), so the occupation design cannot attribute it to AI. A firm-level event study around a firm's first posting for AI-building work finds no detectable change in the working-with-AI duties of its non-technical postings or in the junior share of its other postings (Appendix C.1).

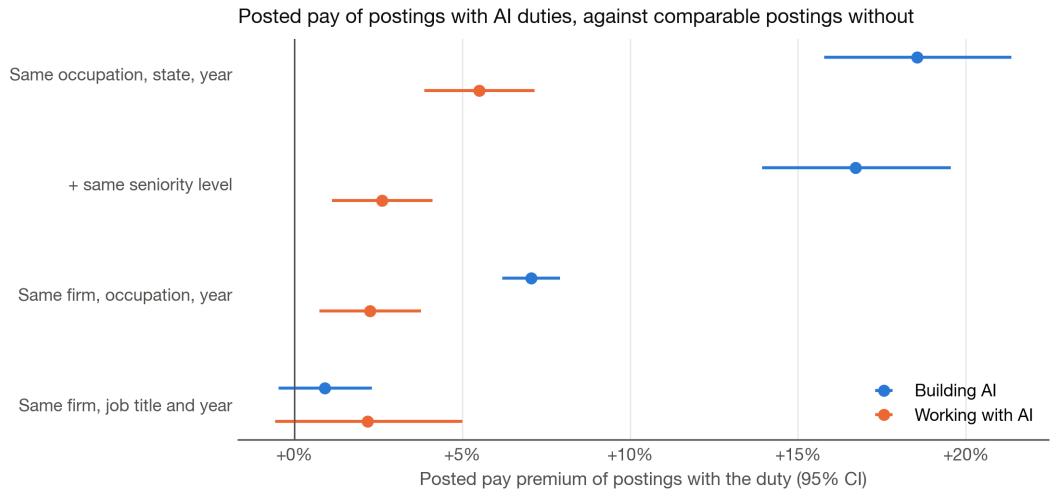


Figure 10: Posted-pay premium of postings with an AI duty, 2023–26, as the comparison tightens: log annual posted pay on indicators for a building-AI and a working-with-AI duty, with fixed effects for the cells named on the left; premia in per cent, $100(e^{\beta}-1)$, with 95% intervals (the text reports log points). Standard errors clustered by firm.

5 Limitations

The postings describe what employers ask for when they hire, not the work that people do, and only for the employers that use these platforms. They over-represent professional, technical and large employers, and their platform mix shifted: Workday postings carry text only from 2022, and Lever's pages stop late in 2025. The comparisons hold when both platforms are dropped, the platform mix is held fixed or the postings are reweighted to the occupation mix of employment or hiring; the levels change with the weights (Appendix Table C1). They also hold in the firms that posted in every year and within firm-by-occupation cells (Appendix D). Which employers the archive captures changed over the period, and nothing is known about employers' exclusions of the crawler after 2023, so the entry and exit of platforms and firms is checked rather than ruled out.

Every label in the paper comes from a language model or from a classifier trained on its labels, and the validation is between models: no human-coded set exists yet. A second model agrees with most of the first's labels and hiding the job title changes them little, but which non-AI lines are new tasks is uncertain: rewording the prompt raises their share by about a third, the second model calls fewer of them new tasks, and new non-AI work written in familiar words passes the screen less often, so the non-AI counts likely understate it and Section 4.3 gives bounds. How the lines are batched for reading also matters, and the paper's levels are those of batches that mix the samples (Appendix B.6). Tools and models released after the language models' training cutoffs are judged from the surrounding context of the duty sentence.

Adverts have grown longer since 2023, and some are now drafted with language models. The embedding screen flags more of their boilerplate as well: benefits and legal text is flagged new 34% of the time in 2026, against 17.4% of duty lines, but the language-model reading sets this material aside, classifying 99% of such lines as non-duties and none as new tasks. New AI duty lines also recur across unrelated firms no more often than continuing duties do, and the headline estimates are unchanged when the largest posters are excluded (Appendix D).

The count of new work asks whether a task is new to the economy's postings, not whether it is new

to the job. Section 4.4 measures one kind of work new to jobs, technical and data tools, on terms fixed in advance; a general measure would need a reliable way to give each duty line a task, which matching lines to O*NET's work activities does not provide (Appendix B.2). A classifier trained on 14,664 lines that Opus labelled with an O*NET work activity gives a rough version: measured against each occupation's own 2019 mix of activities, the share of its duty lines outside that mix did not rise between 2022 and 2026 (Appendix D).

Novelty is measured against one reference period, so the share of new lines grows with time; it is the reading, not the count, that measures new work. The last half-year is partial, and several results sharpen in 2026: without it, the occupation-level effect on working-with-AI duties falls from 1.7 to 1.1 points. Finally, the occupation-level design is descriptive: observed AI use is measured in 2025, it is built on the same task ratings as AI exposure, and occupations that use AI more differ in other ways.

6 Conclusion

Whether a new technology raises the demand for labour depends in part on the new work it creates (Acemoglu and Restrepo, 2018, 2019), and in the past most new work appeared where technologies complemented what workers already did (Lin, 2011; Autor et al., 2024). For generative AI, the evidence has so far come mostly from what the technology could do and what people use it for. Read in job postings, the new tasks are concentrated in the occupations that use it and are mostly about AI itself: first building it, now increasingly using it and driving its adoption, especially outside computer occupations. Where AI is used most, new tasks of other kinds are fewer than new AI tasks, about one against two and a half in every 100 duty lines in 2026, and most of their rise came before ChatGPT. Jobs are also taking up tools that other jobs used before, such as dashboards and workflow automation, more so where AI is used. Kalyani et al. (2025) find in postings that new technologies spread from their pioneers to a wider set of jobs over decades; here the move from building AI to working with it shows up within four years.

These findings carry two broader lessons. First, official occupational task inventories lag behind technological change: nearly four years after ChatGPT, O*NET still lists no task about language models or working with AI, and for only about half of the new AI duties in job postings does the closest task in their occupation's O*NET list describe related work—typically an older analogue such as programming robots or building business-intelligence tools. Pre-existing AI exposure measures built on O*NET (Webb, 2020; Eloundou et al., 2024) still point to the occupations where new AI work is appearing, because new AI tasks are emerging in the same occupations whose existing tasks are exposed to AI, but tracking how work changes within an occupation requires reading what employers actually ask for. Second, the spread of working-with-AI duties shows that employers are hiring workers to use and oversee the technology, but whether these tasks remain a durable part of jobs or prove to be a passing stage before further automation will depend on how far model capabilities advance (Kwa et al., 2025). The posting-level data from this study will be released publicly so that researchers can follow that evolution as generative AI diffuses.

References

Daron Acemoglu and Pascual Restrepo. The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review*, 108(6):

1488–1542, 2018.

Daron Acemoglu and Pascual Restrepo. Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2):3–30, 2019.

Daron Acemoglu, David Autor, Jonathon Hazell, and Pascual Restrepo. Artificial intelligence and jobs: Evidence from online vacancies. *Journal of Labor Economics*, 40(S1):S293–S340, 2022.

Enghin Atalay, Phai Phongthiengtham, Sebastian Sotelo, and Daniel Tannenbaum. The evolution of work in the United States. *American Economic Journal: Applied Economics*, 12(2):1–34, 2020.

David Autor, Caroline Chin, Anna Salomons, and Bryan Seegmiller. New frontiers: The origins and content of new work, 1940–2018. *Quarterly Journal of Economics*, 139(3):1399–1465, 2024.

David H. Autor, Frank Levy, and Richard J. Murnane. The skill content of recent technological change: An empirical exploration. *Quarterly Journal of Economics*, 118(4):1279–1333, 2003.

Alexander Bick, Adam Blandin, and David J. Deming. The rapid adoption of generative AI. Working Paper 32966, National Bureau of Economic Research, 2024.

Erik Brynjolfsson, Bharat Chandar, and Ruyu Chen. Canaries in the coal mine? Six facts about the recent employment effects of artificial intelligence. Working paper, Stanford Digital Economy Lab, 2025a. URL https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/.

Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. *Quarterly Journal of Economics*, 140(2):889–942, 2025b.

Brantly Callaway and Pedro H. C. Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.

Doruk Cengiz, Arindrajit Dube, Attila Lindner, and Ben Zipperer. The effect of minimum wages on low-wage jobs. *Quarterly Journal of Economics*, 134(3):1405–1454, 2019.

Bharat Chandar and Bouke Klein Teeselink. How does AI change labor demand? Evidence from 41 countries. Working paper, Stanford Digital Economy Lab, September 2026.

Wilbur Xinyuan Chen, Suraj Srinivasan, and Saleh Zakerinia. Displacement or complementarity? The labor market impact of generative AI. Working Paper 25-039, Harvard Business School, 2024.

Caroline Chin, Alex Martin Richmond, and OpenAI Economic Research. Work at the frontier: How AI is expanding what people do at work. Report, OpenAI, July 2026.

Common Crawl Foundation. Common Crawl. https://commoncrawl.org, 2026. Main crawls CC-MAIN-2017-04 to CC-MAIN-2026-39.

David Deming and Lisa B. Kahn. Skill requirements across firms and labor markets: Evidence from job postings for professionals. *Journal of Labor Economics*, 36(S1):S337–S369, 2018.

Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702):1306–1308, 2024. doi: 10.1126/science.adj0998.

Edward Felten, Manav Raj, and Robert Seamans. Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12):2195–2217, 2021.

Menaka Hampole, Dimitris Papanikolaou, Lawrence D. W. Schmidt, and Bryan Seegmiller. Artificial intelligence and the labor market. Working Paper 33509, National Bureau of Economic Research, 2025. URL https://www.nber.org/papers/w33509.

Kunal Handa, Alex Tamkin, Miles McCain, Saffron Huang, Esin Durmus, Sarah Heck, Jared Mueller, Jerry Hong, Stuart Ritchie, Tim Belonax, Kevin K. Troy, Dario Amodei, Jared Kaplan, Jack Clark, and Deep Ganguli. Which economic tasks are performed with AI? Evidence from millions of Claude conversations. *arXiv preprint 2503.04761*, 2025.

Brad Hershbein and Lisa B. Kahn. Do recessions accelerate routine-biased technological change? Evidence from vacancy postings. *American Economic Review*, 108(7):1737–1772, 2018. doi: 10.1257/aer.20161570.

Seyed M. Hosseini and Guy Lichtinger. Generative AI as seniority-biased technological change: Evidence from U.S. résumé and job posting data. Working paper, SSRN 5425555 (first version August 2025), 2026.

Anders Humlum and Emilie Vestergaard. Large language models, small labor market effects. Working Paper 33777, National Bureau of Economic Research, 2025. URL https://www.nber.org/papers/w33777.

Zanna Iscenko and Fabien Curto Millet. Looking for the ladder: Is AI impacting entry-level jobs? Economic Innovation Group, January 2026.

Aakash Kalyani, Nicholas Bloom, Marcela Carvalho, Tarek A. Hassan, Josh Lerner, and Ahmed Tahoun. The diffusion of new technologies. *Quarterly Journal of Economics*, 140(2):1299–1365, 2025.

David Koll, Elmer Z. Li, and Marina M. Tavares. New skills, higher pay? Evidence from job postings. Working paper, February 2026.

Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Chris Painter, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long software tasks. *arXiv preprint 2503.14499*, 2025.

Jeffrey Lin. Technological adaptation, cities, and new work. *Review of Economics and Statistics*, 93(2):554–574, 2011. doi: 10.1162/REST_a_00079.

David Marguerit. Augmenting or automating labor? The effect of AI development on new work, employment, and wages. arXiv preprint 2503.19159, 2025.

Nestor Maslej, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Njenga Kariuki, Emily Capstick, Anka Reuel, Erik Brynjolfsson, John Etchemendy, Katrina Ligett, Terah Lyons, James Manyika, Juan Carlos Niebles, Yoav Shoham, Russell Wald, Tobi Walsh, Armin Hamrah, Lapo Santarlasci, Julia Betts Lotufo, Alexandra Rome, Andrew Shi, and Sukrut Oak. The AI

Index 2025 annual report. Technical report, Stanford Institute for Human-Centered Artificial Intelligence, 2025.

Maxim Massenkoff and Peter McCrory. Labor market impacts of AI: A new measure and early evidence. Anthropic research report, March 2026. URL https://www.anthropic.com/research/labor-market-impacts.

Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. arXiv preprint 1802.03426, 2018.

Stephen Meisenbacher, Svetlozar Nestorov, and Peter Norlander. Extracting O*NET features from the NLx corpus to build public use aggregate labor market data. arXiv preprint 2510.01470, 2025.

Shakked Noy and Whitney Zhang. Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192, 2023.

Kelsey Purcell. Applicant tracking system (ATS) usage report, 6th edition. Jobscan, 2026. URL https://www.jobscan.co/blog/fortune-500-use-applicant-tracking-systems/.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992, 2019.

Soumitra Shukla. Technology bottlenecks: Measuring what AI cannot do inside jobs. Working paper, Burning Glass Institute, October 2026.

Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.

Kiran Tomlinson, Sonia Jaffe, Will Wang, Scott Counts, and Siddharth Suri. Working with AI: Measuring the applicability of generative AI to occupations. arXiv preprint 2507.07935, 2025. URL https://arxiv.org/abs/2507.07935.

Fangyan Wang, Zaiyan Wei, and Yang Wang. Generative AI and the reorganization of labor demand. arXiv preprint 2605.23159, 2026.

Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

Michael Webb. The impact of artificial intelligence on the labor market. Working paper, Stanford University, 2020.

A The dataset

This appendix summarises the construction and quality of the postings. The build covers January 2017 to September 2026; this paper uses postings dated from 2019. Fuller documentation, including the extraction rules and the occupation classifier's training labels (posting identifiers and codes), will accompany the release.

A.1 Construction

Table A1 lists the platforms and Table A2 accounts for every captured page. Table A3 gives counts and field completeness by year. Counts follow Common Crawl's crawling, not the labour market: from September 2017 to January 2019 the archive captured Workday and iCIMS pages at several times their later rate, iCIMS pages before September 2017 carry no text, and 2023 had fewer crawls than the years around it. I therefore start the analysis in 2019 and do not interpret posting counts as vacancies. The crawl-intensity weight of Table A3 is the inverse of a crawl's capture intensity for the platform, relative to the platform's 2019–20 rate, where that intensity exceeds one; it differs from one only in the crawls of 2017 to January 2019, so no result in this paper uses it, and Appendix B shows the measure without the January 2019 crawl instead. Table A4 follows the postings dated 2019–26 to the analysis sample.

Table A1: Platforms

Platform	US ATS pages (%)	US postings	Years	Description (%)	Salary (%)	State (%)	Verdict
iCIMS	36.2	1,393,482	2017–2026	88	17	84	yes
Workday	20.2	634,446	2017–2026	26	11	66	caveats
Greenhouse	10.3	446,489	2017–2026	100	44	83	yes
Jobvite	4.1	171,110	2017–2026	100	21	85	yes
JazzHR	3.2	96,274	2017–2026	84	30	78	caveats
Breezy HR	2.6	99,834	2017–2026	95	30	87	yes
Lever	2.5	117,637	2017–2026	100	31	71	caveats
UKG (UltiPro)	2.3	69,272	2017–2026	76	16	74	caveats
SmartRecruiters	2.2	83,018	2017–2026	78	12	83	caveats
Dayforce	1.9	76,437	2019–2026	97	23	93	yes
Paylocity	1.3	56,490	2017–2026	100	33	93	yes
Ashby	1.2	51,563	2020–2026	79	50	55	caveats
Paycom	1.0	38,143	2019–2026	93	37	86	caveats
Recruitee	0.3	7,677	2018–2026	94	36	91	caveats

Notes: US ATS pages: the platform's share of the US job pages captured on the 20 platforms first queried, in one crawl per quarter from 2017 to 2026. US postings: postings in the release. Description, salary, state: share of the platform's US postings with a description of at least 200 characters, salary information and a resolvable state.

Table A2: From captured pages to postings, 2017–2026

Step	All	US	Non-US	Unresolved
Captured job pages on the 14 platforms	10,505,800			
not fetched or not parsed (kept)	3			
Jobs (platform, employer and job ID)	5,205,625	3,630,815	1,348,062	226,748
repost merge (same title, location and text)	−350,263	−285,346	−63,262	−1,655
Postings	4,855,362	3,345,469	1,284,800	225,093
non-US postings dropped	−1,284,800		−1,284,800	
Postings kept (US and unresolved)	3,570,562	3,345,469		225,093
iCIMS portal merge	−4,163	−3,597		−566
Release	3,566,399	3,341,872		224,527
of which: closed-ad notice (flagged)	34,995			
of which: no title	24,232			
of which: no description of 200+ characters	888,023			

Notes: Counts at each step of the build, which covers 2017–2026. Jobs are called US, non-US or unresolved before the repost merge, postings after it.

Table A3: US postings by first-seen year

		Weighted		Field present (%)						Median
Year	Postings	postings	Employers	Description	Duties	Date	Employer	State	Salary	length
2019	274,531	225,538	10,739	87	69	80	75	91	6	3,142
2020	197,526	197,526	10,311	85	64	84	75	88	7	3,217
2021	308,332	308,332	12,408	88	70	87	84	88	13	3,456
2022	287,957	287,957	13,161	87	69	85	78	81	25	3,594
2023	192,502	192,502	12,606	91	70	88	82	83	35	3,691
2024	326,187	326,187	16,624	90	68	87	84	82	42	4,045
2025	371,142	371,142	18,314	88	69	85	79	78	48	4,551
2026	327,003	327,003	16,065	87	69	84	79	75	51	4,967

Notes: Weighted postings use the crawl-intensity weight, which scales crawls that captured a platform at an unusual rate to the platform's 2019–20 capture rate. Description: at least 200 characters. Duties: a duties section was found by header (the text layer of Section 2 replaces this rule). 2026: January to September.

Table A4: From the release to the analysis sample, 2019–26

Step	Postings
release, dated 2019–26 (US or unresolved location)	2,390,292
US location	2,256,069
(the steps below keep unresolved locations)	
English, description of at least 200 characters	2,009,281
not a bulk poster	1,929,848
one posting per location group	1,712,526
at least one duty line (analysis sample)	1,629,317
of which: location unresolved	28,835
of which: recruitment agency	17,338
of which: closed-advert notice	50

Notes: Postings of 2019–26, dated as in Section 2, at each step of the filters of Section 2. The last three rows are flags inside the analysis sample, which keeps those postings.

A.2 Coverage and representativeness

Figure A1 summarises the benchmarks. Where and when states required pay ranges in postings, the share of postings with pay information jumped (panel a); the share offering remote or hybrid work rose with the pandemic, as in Lightcast's and Indeed's series, if to a higher level (panel b); posting shares by detailed occupation track Lightcast's (panel c); and median posted pay tracks the occupation's median wage (panel d). Figure A2 compares the postings with hires, Figure A3 with the occupation mix of employment and of other posting data, and Table A5 summarises the differences as dissimilarity indices: the share of postings that would have to change category for two distributions to match. Figure A4 compares states, and Table A6 the shares of postings that state pay or offer remote work against Indeed's and Lightcast's series. Live postings per crawl fell by 39% between February and May 2020 and track the Indeed postings index until 2023 but not after (Figure A5), as the share of each platform that the archive captured changed. Table A7 compares extracted fields with Web Data Commons' independent parse of the same pages.

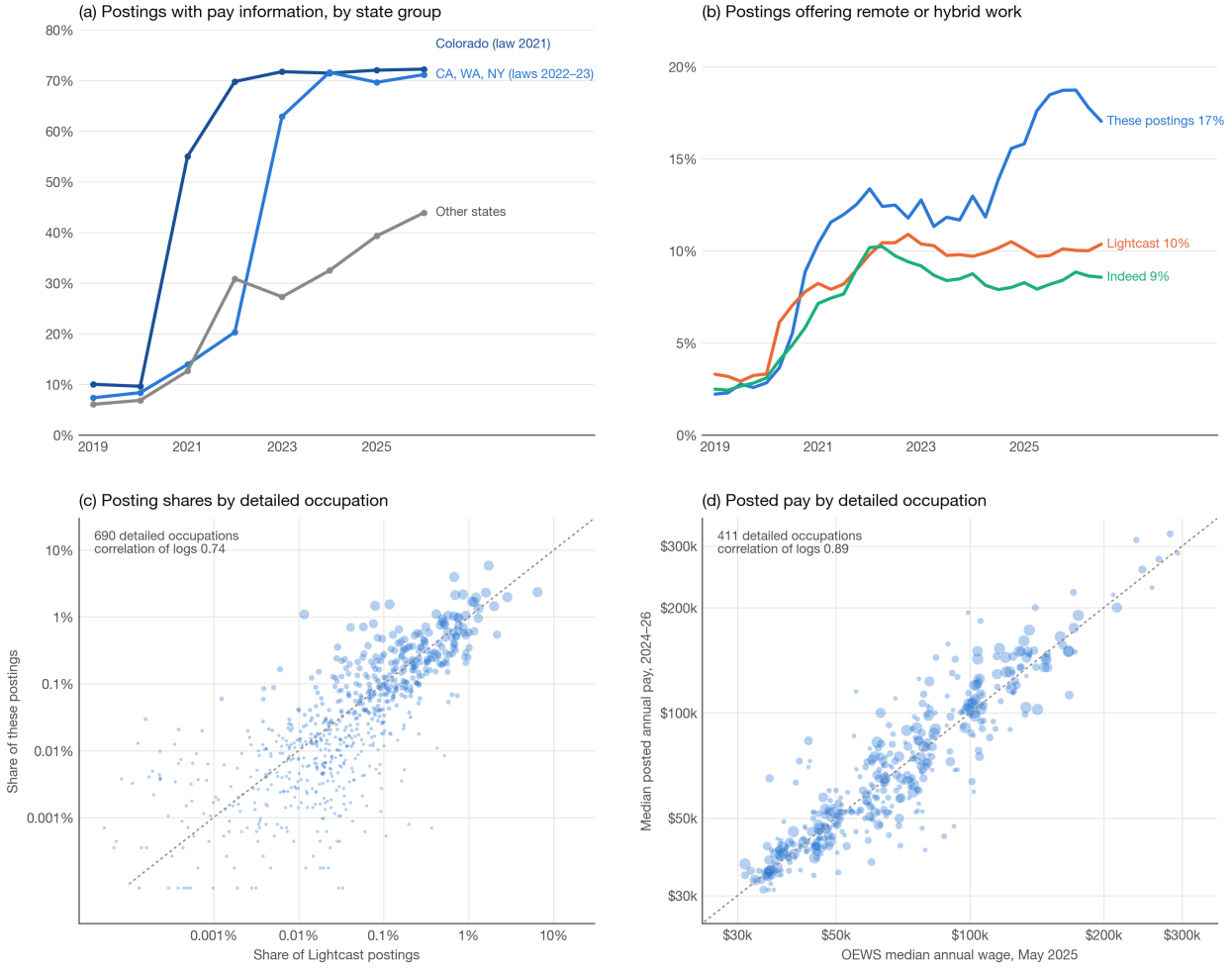


Figure A1: The postings against known facts. (a) Share of postings with pay information by state group; Colorado's pay-transparency law took effect in 2021, California's, Washington's and New York's in 2022–23. (b) Share of postings offering remote or hybrid work, by quarter, against Lightcast's and Indeed's series. (c) Shares of postings by detailed occupation against Lightcast's, 2023–26 (log scales). (d) Median posted annual pay, 2024–26, against the OEWS median wage by detailed occupation (log scales). Dotted lines: equality.

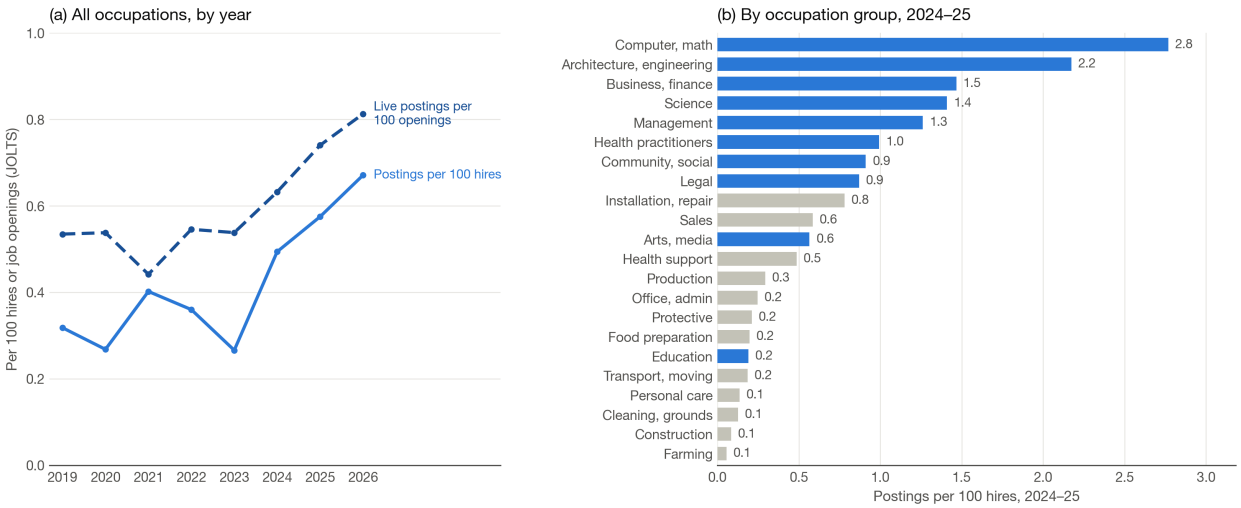


Figure A2: Coverage of US hiring. (a) Postings per 100 hires in JOLTS, and live postings per crawl per 100 job openings. (b) Postings per 100 hires by occupation group, 2024–25; blue: professional and technical groups.

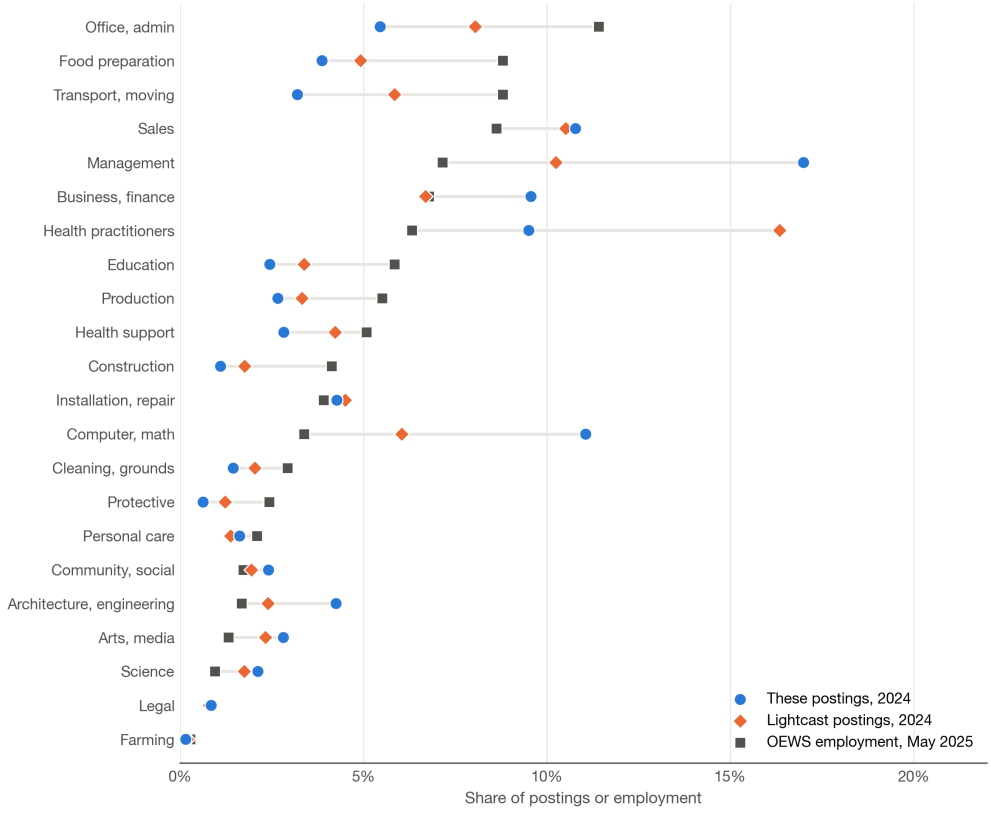


Figure A3: Occupation mix by SOC major group: these postings and Lightcast postings in 2024, and OEWS employment in May 2025.

Table A5: Dissimilarity indices

Comparison	Level	ATS postings weighted by		
		crawl intensity	OEWS	hires
ATS vs OEWS employment	major group	0.34	0.00	0.14
	detailed	0.45	0.32	0.35
ATS vs Lightcast postings	major group	0.22	0.21	0.28
	detailed	0.38	0.38	0.42
Lightcast vs OEWS	major group	0.21		
	detailed	0.35		
ATS vs CES employment, 2024	state	0.12		
ATS vs JOLTS openings, 2024	state	0.13		

Notes: Share of postings that would have to change category for the two distributions to match. Occupations: postings first seen in 2023–26; OEWS employment in May 2025. Columns: crawl-intensity weight, and the representativeness weights that give each major group its share of OEWS employment or of new hires in the CPS (the OEWS weights set the major-group index against OEWS to zero by construction). States: 2024.

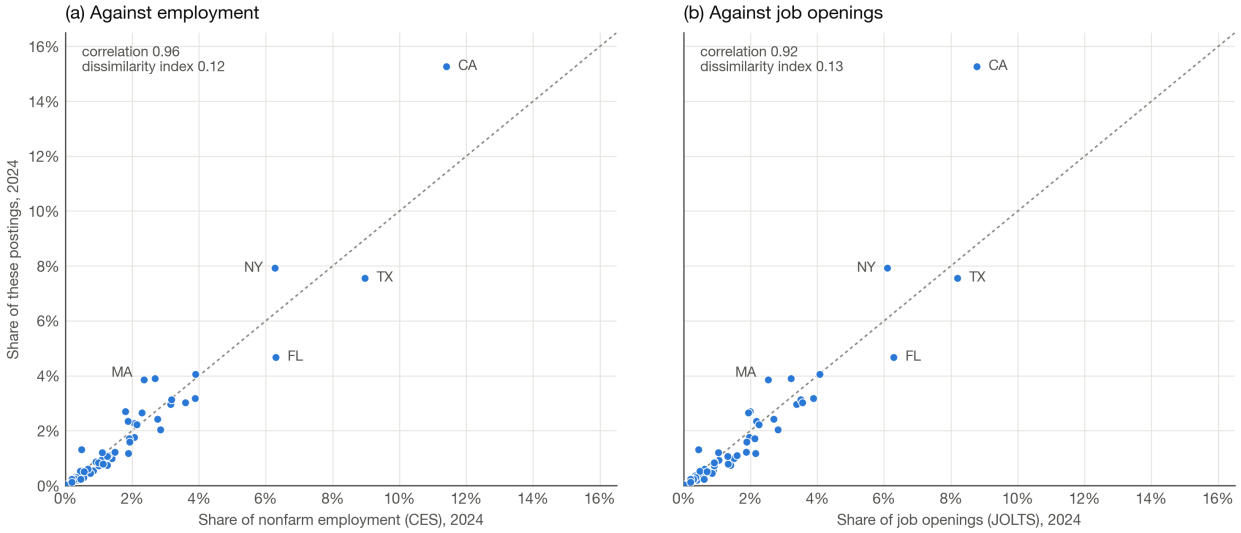


Figure A4: State shares of postings against state shares of nonfarm employment (a) and of job openings (b), 2024. Dotted lines: equality.

Table A6: Fit of the pay-information and remote-work series

Series	Period	Quarters	Correlation			Gap (points)		
			levels	changes	changes[a]	2021	2023	2025
Pay information vs Indeed	2020–26	27	0.98	0.24	0.30	−11.9	−6.0	−1.7
postings with a description	2020–26	27	0.98	0.19	0.32	−10.0	−2.1	5.0
Remote work vs Lightcast	2019–26	31	0.88	0.14	0.33	3.3	1.8	7.7
Remote work vs Indeed	2019–26	31	0.85	0.48	0.33	3.8	3.2	9.5

Notes: Quarterly shares of postings against Indeed's pay-transparency and remote-work trackers and Lightcast's remote-work share. Changes: first differences. [a]Postings dated by their date posted where available.

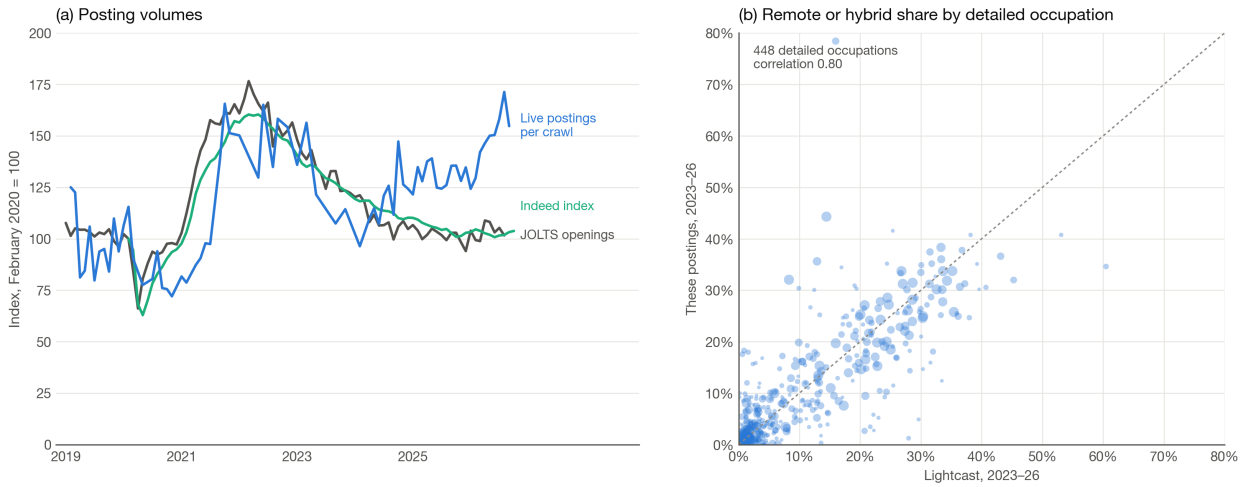


Figure A5: (a) Live US postings per crawl, JOLTS job openings and the Indeed postings index (February 2020 = 100). (b) Share of postings offering remote or hybrid work by detailed occupation, against Lightcast, 2023–26. Dotted line: equality.

Table A7: Agreement with Web Data Commons

Field	Route	Matched pairs	Compared	Agree (%)
title	URL (Workday)	2,628	7	
	page template	7,577	7,543	99.4
	structured data	102,617	102,579	100.0
	all	112,822	110,129	99.9
US call	metadata or description	1,954	3	
	page template	9,918	7,486	100.0
	structured data	100,950	100,855	100.0
	all	112,822	108,344	100.0
location	URL (Workday)	2,628	1,104	99.7
	page template	7,577	7,497	97.4
	structured data	102,617	97,036	97.2
	all	112,822	105,637	97.2
description	page template	8,489	6,525	63.8
	structured data	104,333	104,321	100.0
	all	112,822	110,846	97.8

Notes: Web Data Commons postings on the fourteen platforms in the October 2022 and October 2024 crawls, matched by platform, employer and job identifier. Route: where the field came from in this build.

A.3 Occupation codes

The classifier blends a linear model of the title's words and characters and the posting's duty and role sentences with a small language model (MiniLM) fine-tuned on the same text. It is trained on 63,000 postings coded by Opus 5.5, 60,000 drawn by platform and year and 3,000 with AI or new titles, and on postings whose titles match an O*NET title. On 5,179 held-out postings drawn by platform and year it agrees with Opus 5.5 on 74% of detailed codes and 87% of major groups; Sonnet 5.5 and Opus 5.5 agree with each other on 73%, and Opus with itself on a second reading on 89%. On 800 postings with AI or new titles it agrees on 78%. A code is confident when the classifier's probability is at least 0.55, where its codes agree with Opus as often as Opus agrees with itself: 64% of postings, at 90% agreement, and 89% on AI and new titles. The right code

is among the three likeliest for 89% of postings. Two parts of the pipeline use title-based codes from a simpler classifier that reads only the title and the first 400 characters of the description: the strata from which the reading samples are drawn (Appendix B.6) and the themes (Table C5); because these codes rest on the title rather than the duties, they also serve as a check (Appendix D). Figure A6 shows that the classifier is conservative, with agreement at or above its confidence, and how agreement rises with the threshold. Confident codes cover 64–68% of postings in every year, and 79% of computer postings, 85% of health-care postings, 43% of production postings and 47% of science postings.

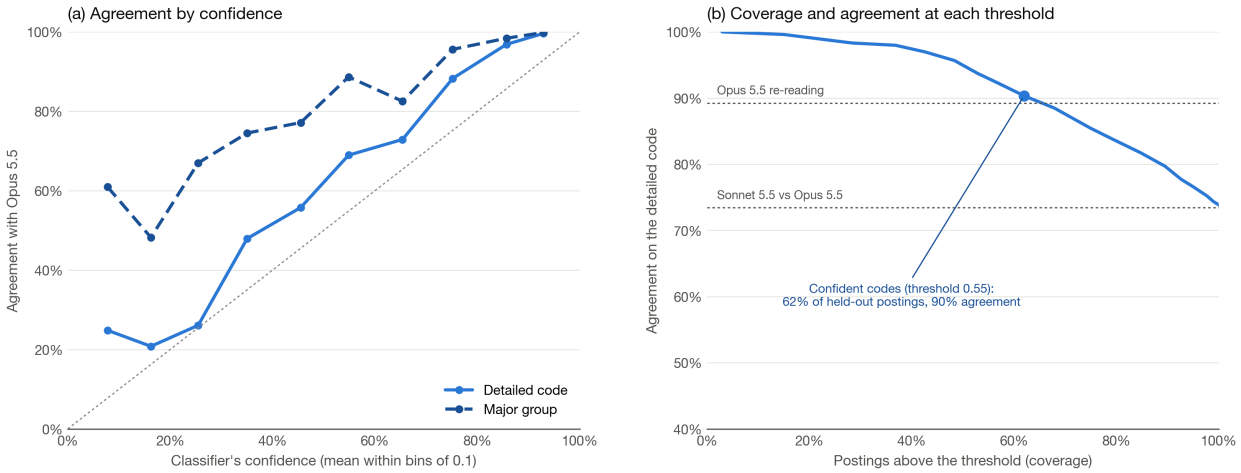


Figure A6: (a) Agreement with Opus 5.5 by bin of 0.1 of the classifier's confidence. (b) Share of held-out postings above each confidence threshold and agreement on them; dot: the threshold for a confident code (0.55). Dotted lines: Sonnet 5.5 against Opus 5.5, and Opus against its own second reading.

A.4 The text layer

The duty classifier is a regularised logistic regression on the sentence's words and character sequences, its section header, the type of section, its position in the advert and, in a second stage, the predicted classes of the sentences around it. Typing a sentence is a narrow task, carried mostly by its wording, its header and where it sits, so a linear model learns it well from the 45,000 labelled sentences of 1,200 postings, 1,000 drawn by year and 200 with flat text; a retest of 100 postings by Sonnet agrees on 99% of duty labels. Table A8 reports the classifier's agreement with the labels by year, in five-fold cross-validation that holds out whole postings, so each score is on adverts the classifier was not trained on. The weakest class is statements about the role, a fifth of which the classifier calls duties. Flat text is restored to lines by rules before the sentences are classified. Flattening 4,061 duty sentences of pages that have native line breaks and restoring them recovers 56% exactly, and most of the rest merged with a neighbour; merged lines sit farther from any single 2019 duty and are flagged new more often (19% against 12% of the original lines), which is why Appendix D repeats the comparisons without the flat-text platforms. In 2026, 52% of AI duty lines and 62% of other duty lines sit under a duties header, and 22% of AI lines under a header about the role, where statements about the role pass the classifier; the reading of the new lines sets such statements aside.

Table A8: Duty classifier against language-model labels

Year	Postings	Units	Precision	Recall	Duty share, Sonnet (%)	Predicted (%)
2017	100	3,067	0.90	0.94	32.0	33.3
2018	100	3,359	0.92	0.95	30.9	31.9
2019	100	3,525	0.92	0.93	28.3	28.7
2020	100	3,669	0.92	0.95	30.9	32.0
2021	100	3,351	0.94	0.93	27.9	27.8
2022	100	3,927	0.92	0.92	20.0	20.0
2023	100	3,799	0.94	0.94	24.1	24.0
2024	100	3,694	0.92	0.96	26.7	27.7
2025	100	4,379	0.91	0.96	23.7	25.1
2026	100	4,695	0.89	0.94	23.2	24.3
All (weighted)	1,000	37,465	0.92	0.94	26.4	27.1
Flat text	283	10,148	0.92	0.93	21.3	21.5

Notes: Five-fold cross-validation grouped by posting. Units: sentences and headers. Precision: share of predicted duties that the language model labels duties; recall: share of the language model's duties predicted. Duty share: share of units labelled or predicted duty.

A.5 Known gaps

Workday pages carry descriptions only where they include structured data, almost none before 2022 and about half from 2023, and their titles and locations come from the page address. Common Crawl holds almost no UKG pages in 2021 and almost no Lever or Paycom postings in 2026. Four tested platforms whose archived pages mostly carry no job data, Taleo, ADP, Workable and BambooHR, are not included, nor are ATS platforms outside the 20 first queried, and school districts, public employers and health systems mostly post elsewhere.

B Measurement

B.1 Novelty and the COVID-19 check

The embedding model is all-MiniLM-L6-v2 (mean pooling, 128 tokens, unit-normalised vectors). The search is exact: every distinct duty sentence of 2019H2–2026 is compared with every distinct duty sentence of 2019H1. Figure B1 shows what the similarity means. Sonnet 5.5 read 100 duty lines of 2023–26 in each bin of 0.05 of similarity to their most similar duty of 2019H1, from 0.45 to 0.95, beside that duty, with the prompt of the new-line reading (Appendix B.6). Re-wordings dominate above 0.75, variants between 0.55 and 0.75, and new tasks and lines that are not duties are most common in the most distant pairs. To measure what the screen misses, Sonnet also read 400 non-AI lines in each of three bins just above the threshold (0.66–0.70, 0.70–0.75 and 0.75–0.80) and called 2 of the 1,498 lines it read above the threshold new tasks. Weighting each bin's rate of new tasks by its lines in 2023–26, the screen passes 87% of the non-AI lines Sonnet would call new tasks. Most lines sit in the bins above the threshold, and a rate of zero in a sample of 400 is uncertain: drawing each bin's rate from its sampling distribution gives a median of 73% and a 95% interval of 47–91%.

Table B1 applies the screen and the reading to new work that is known to have arrived. Of the COVID-19 duty lines, the screen flags 67.7% and Sonnet reads 38% of these as new tasks; another

32% are notices about COVID-19 policy that the text layer took for duties. For four kinds of non-AI work that are new, or much more common, since 2019, Sonnet reads 20–72% of the flagged lines as new tasks, but the screen flags only 26–49% of the lines, and some of the rest are new tasks written in familiar words: by these readings the screen passes 48–69% of the new tasks.

Table B1: The screen and the reading on known new work

	Lines read	Flagged new (%)	New tasks, flagged (%)	New tasks, not flagged (%)	Recall (%)
COVID-19 duties (2020–22)	800	68	38	12	88
Climate disclosure (CSRD, ISSB, TCFD, SEC)	84	48	52	52	48
Electric-vehicle batteries and charging	400	49	26	11	69
New compliance regimes (CMMC, SBOMs, No Surprises Act, SOFR, T+1)	400	44	72	39	59
Telehealth	400	26	20	7	49

Notes: Duty lines of 2023–26 (COVID-19: 2020–22) that contain the named terms and are not about AI. Flagged new: share below the novelty threshold (COVID-19: of 2020–21 duty lines). New tasks: share of the flagged and of the other lines that Sonnet 5.5 reads as new tasks, each beside its most similar 2019H1 duty (400 lines of each kind, flagged or not; COVID-19: 400 flagged and 400 not; climate disclosure: all 84 lines). Recall: share of the lines read as new tasks that the screen flags.

Dropping January 2019 from the reference changes little. The last crawl of the kind that captured some platforms several times over falls in January 2019, and 23% of the reference sentences appear only in postings first captured then; without them, the threshold that makes 10% of 2019H2 new is 0.652 rather than 0.66, the levels of new AI lines fall slightly (Table C1), and genuinely new work in the high-use third goes from 0.75 to 3.24 per 100 duty lines, against 0.7 to 3.5 (Appendix D). Measured against the duties of the two previous years instead, on 20,000 lines a year, the result depends on the size of the reference, since a larger reference finds closer matches: the full two-year windows grow from 2.1 to 3.7 million distinct sentences, and against them new non-AI lines fall from 5.5 per 100 duty lines in 2021 to 4.0 in 2026. Against windows subsampled to the size of the 2019H1 reference (679,314 sentences), new non-AI lines stay at 11.5 to 11.7 per 100 over 2021–26 while new AI lines rise from 0.13 to 0.62: non-AI novelty did not accelerate, and the moving reference counts each year's new wording rather than all change since 2019. A larger embedding model (BAAI bge-base), with its own threshold for the same 10% churn (0.77), gives the same series on samples of 20,000 lines per half-year: 1.5 new AI lines in every 100 duty lines in 2026, against 1.8 with MiniLM on the same lines, and the same contrast between the thirds (Table C1). The two models agree on the totals more than on single lines: each flags 58–71% of the lines the other flags.

Figure B3 shows the COVID-19 check. Sentences of all kinds with a COVID-19 term rose in 2020 and peaked in 2022, but most were policy notices, legal text and statements about the company; duty lines with a COVID-19 term peaked earlier, at 0.21% of all duty lines in the first half of 2021 (panel a); in 2021, 11% of sentences with a COVID-19 term were duty lines. COVID-19 duty lines were most common in the occupations that use AI least: in 2021H1 they were 0.32 in every 100 duty lines of the third with the least AI use and 0.12 in the third with the most. The themes of the new lines of 2020–21, which include the policy notices, show both kinds (panel b): four rising themes are COVID-19 duties and eight are notices. Figure B2 sets the new AI lines of 2023–26 against the COVID-19 duty lines of 2020–22: across all occupations, new AI duty lines in 2026 are

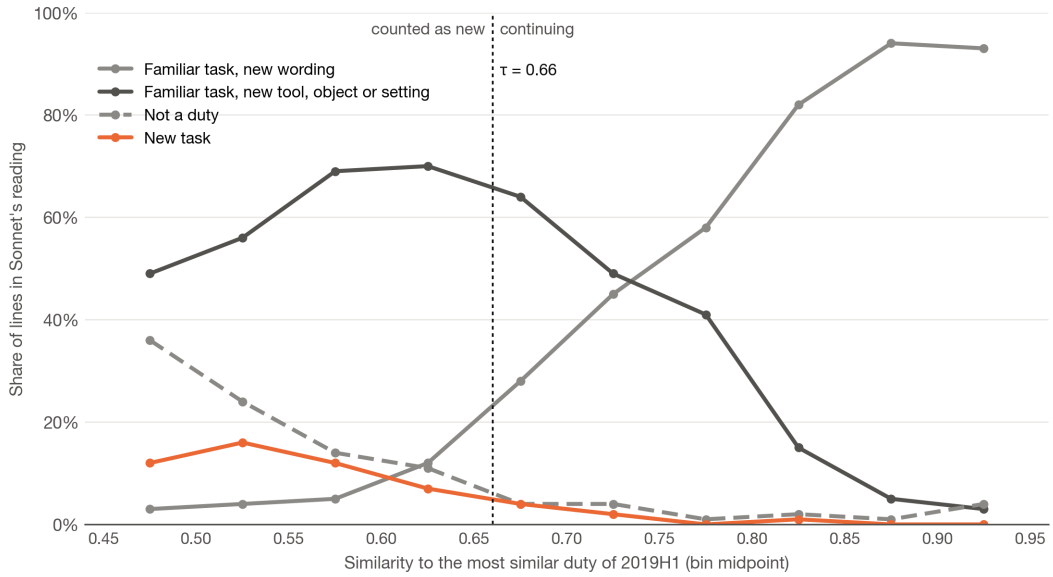


Figure B1: What similarity to 2019 means: Sonnet 5.5's reading of 1,000 duty lines of 2023–26, each beside its most similar duty of 2019H1, 100 per bin of similarity. Dotted line: the novelty threshold.

8 times as common as COVID-19 duty lines were at their 2021H1 peak (21 times after reading).

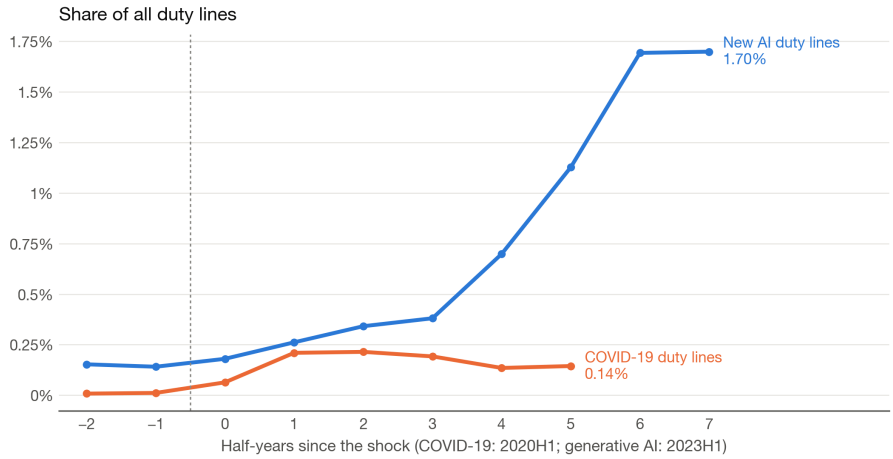


Figure B2: New AI duty lines and COVID-19 duty lines as a share of all duty lines, in half-years since each shock (COVID-19: 2020H1; generative AI: 2023H1). COVID-19 duty lines: duty lines with a COVID-19 term; notices of COVID-19 policy, such as vaccination requirements, are not duty lines and are not counted. COVID-19 series to 2022H2.

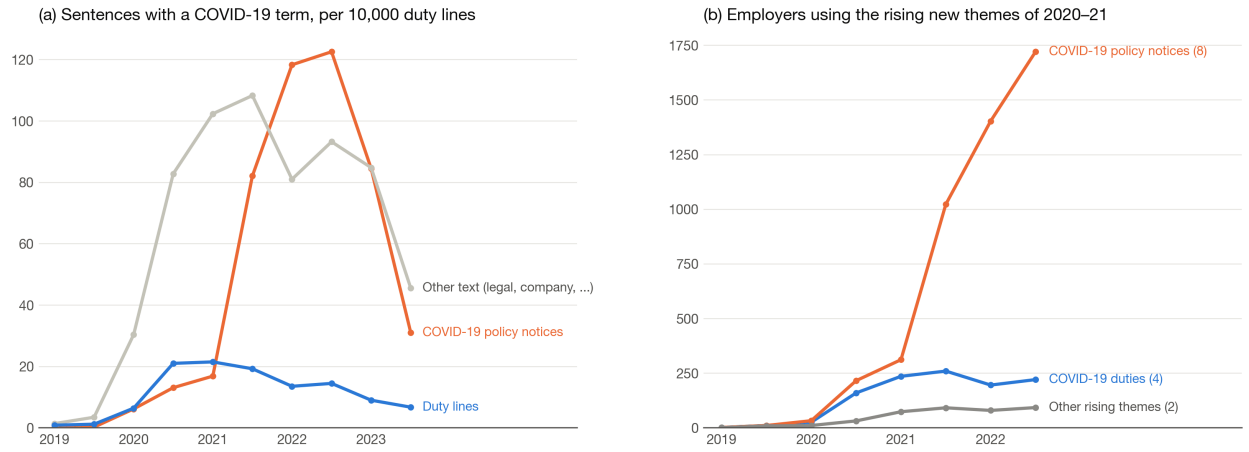


Figure B3: The novelty measure finds COVID-19. (a) Sentences with a COVID-19 term per 10,000 duty lines, by the text layer's class: duty lines, COVID-19 policy notices and other text. (b) Employers using the rising new themes of 2020–21, by whether Sonnet 5.5 judged the theme to be COVID-19 duties, COVID-19 notices or neither. Themes: clusters of the new lines and notices of 2020–21, rising when used by at least three times as many employers as in 2019H2.

B.2 Tools new to the job

Tasks new to the job are hard to measure in general. Mapping each duty line to one of O*NET's 2,087 detailed work activities, through its closest task statement, gives a task to almost every line but not a reliable one: of the pairs that the paper's reading calls a familiar task in new words, only 21–27% get the same work activity (or the same one of O*NET's 332 broader activities), and against a 2019–20 baseline 16–68% of the postings of 2021, before generative AI, would already ask for a task new to their job, depending on how strictly lines are matched. The measure of Section 4.4 is narrower and defined in advance.

Table B2 lists the five families. Each follows O*NET 27.1's Technology Skills categories (November 2022). Its tools are the examples that O*NET flags as hot technologies, those most often requested in postings (47 tools after 13 are dropped as not data or code tools, such as LinkedIn and cloud infrastructure, as office software, Microsoft Access, or as ambiguous words, such as C, R, Go and Swift), and 32 current tools that O*NET 27.1 lacks, such as Power BI, Looker, Snowflake and Zapier, with the words dashboard and data visualisation. Workflow automation has no O*NET category and is defined by its tools alone. Matching is by regular expression on each duty line and, separately, each requirement sentence. The list was fixed before counting and revised once, after a first check by Opus 5.5 of 1,440 matched lines found two systematic false matches: tool vendors describing their own products, and SQL meaning a sales-qualified lead in marketing postings. A tool does not count in a posting by its own vendor, and SQL does not count next to MQL, sales-qualified or SQLs.

A second check, on a fresh sample of 1,440 lines, 80 in each of 18 cells of family, period (2019–22 or 2025–26) and whether the occupation used the family before, measures how often a matched duty asks the worker to use the tool. Where the occupation used the family before, 91–100% do; where it did not, 56–94%, lowest for databases, where most of the remaining false matches are partner firms that sell alongside data platforms. The series weights each matched posting by its family's precision in its period (2023–24: the mean of the two), and its interval comes from 200 replicates that resample employers and draw each precision from its validation sample. Uncorrected, the

share goes from 0.49% in 2022 to 1.46% in 2026; counted in requirement sentences instead of duties, from 0.78% to 1.54%. By family, uncorrected, dashboards and business intelligence rose from 0.33% to 1.21%, workflow automation from 0.16% to 0.50%, databases from 0.09% to 0.30% and programming from 0.07% to 0.20%; statistical software did not rise.

The cutoff and the baseline matter little. Non-use defined at 0.5% or 2% of 2019–20 postings gives 0.89% and 1.83% in 2026 (0.22% and 0.60% in 2022); occupations with at least 1,000 or 2,000 postings in 2019–21 instead of 500 give 1.28% and 1.20%; without Workday the share goes from 0.35% to 1.30%, and in the firms that posted in every year 2022–26 from 0.32% to 1.05%; counting a tool named in the requirements of 2019–20 as prior use lowers the level, to 0.18% in 2022 and 0.79% in 2026, and the rise by about a third. Defining non-use on 2019 alone and counting new tools in 2020–21, before generative AI, adds 0.11 points in two years, against 0.88 from 2022 to 2026. Of the matched duty lines of 2025–26 in occupations that did not use the family, 36% also pass the novelty screen. The precision of the matches is 73% in the third that uses AI most, 79% in the middle third and 78% in the third that uses it least, so the uniform weights overstate the gradient slightly; among the confirmed lines the work is data analysis (18%), automating workflows (17%), dashboards and reports (16%) and code (13%). The gap in working-with-AI duties between these postings and others in the same occupations is 12 points, and 10 within the same firm and occupation.

Table B2: Families of technical and data tools

Family	O*NET categories	O*NET tools (first eight)	Added	New, 2019–22	New, 2025–26	Earlier use
Business intelligence and dashboards	Business intelligence and data analysis; data base reporting	Alteryx, MicroStrategy, SQL Server Reporting Services SSRS, QlikView, Tableau	Power BI, Looker / Looker Studio, Google Data Studio, Domo, Sisense, ThoughtSpot	76	94	91
Databases and data platforms	Data base user interface and query; data base management system	DynamoDB, Redshift, Cassandra, Hadoop, Hive, Spark, Elasticsearch, GraphQL	Snowflake, BigQuery, Databricks, PostgreSQL, MySQL	60	56	91
Programming and scripting	Object or component oriented development; development environment	Kafka, Subversion SVN, C#, C++, Eclipse IDE, .NET Framework, PowerShell, Team Foundation Server	JavaScript, Jupyter, pandas, Google Apps Script	74	85	91
Statistical and analytical software	Analytical or scientific	ArcGIS, SPSS Statistics, SAS, MATLAB	Stata, Minitab, JMP	76	84	100
Workflow automation	(none)	—	Zapier, Power Automate / Microsoft Flow, Power Apps, UiPath, Automation Anywhere, Blue Prism	70	84	–

Notes: O*NET categories: O*NET 27.1 Technology Skills commodity categories. O*NET tools: hot-technology examples kept, the first eight alphabetically, vendor names shortened. Added: current tools that O*NET 27.1 lacks, the first six. Last three columns: precision (%), the share of matched duty lines that Opus 5.5 reads as asking the worker to use, build or work with the tool, in occupations that did not use the family in 2019–20 (new), in 2019–22 and 2025–26, and in those that did (earlier use), in 2025–26; 80 lines each.

B.3 AI labels

On 2,000 AI sentences drawn at random and rated blind by both models, Opus 5.5 and Sonnet 5.5 agree on whether the sentence describes AI work in 98% of cases, on the side in 90% (Cohen's κ 0.82) and on the activity in 85%, and they give the same aggregate shares. AI work without an AI keyword is not rare: Sonnet finds it in 0.6% of a random sample of 139,346 duty sentences without a keyword (18,750 lines a year) and in 8.5% of sentences with near-miss words such as "agent", "automation" or "model". Table B5 scales the AI rate of the random sample to all keyword-free lines: the keywords catch 37–55% of building-AI work, depending on the year, but 87–88% of working-with-AI work in 2025–26. The classifier that extends the labels to every duty line agrees with Sonnet on the labelled sentences (precision 0.98, recall 0.95) but finds few of the AI sentences without keywords (recall 0.16). The paper therefore measures the mix of sides on the keyword lines, scaled by each side's catch rate in each year (Table B4); counting the keyword lines alone, or every duty line with the classifier's labels, gives a larger share of working with AI in 2026 (30% and 25%) and the same rise. Read without the job title and occupation group, 82–93% of AI sentences keep their side, and the share of working with AI in each occupation group moves by at most 1 point.

In symbols, with k_s the keyword lines of side s in a year and c_s the share of the side's work that the keywords catch, estimated as $k_s/(k_s + \hat{n}_s)$ where $\hat{n}_s$ is the side's AI work among keyword-free lines, scaled up from the random sample and from a second stratum of lines with near-miss words, the corrected share of working with AI is $(k_u/c_u)/\sum_s k_s/c_s$, summing over building, working with, and governing or other AI. The catch rate for working with AI in 2022 rests on 5 keyword-free hits (62%, interval 29–85%), so the 2022 share alone has an interval of 3–8%; pooled over 2019–23, when the share was flat, it is 4% (3.8–6.0%), the figure the text uses, and in 2026 it is 22% (20.7–22.8%), with catch rates of 51% for building AI and 86% for working with it. Table B3 lists the keyword and near-miss rules.

Table B3: The AI keyword and near-miss rules

Rule	Patterns (regular expressions, word-bounded)
Any case	artificial intelligence; machine[-]learning; deep[-]learning; neural networks?; large language models?; language models?; foundation models?; generative ai; gen[-]?ai; chatgpt; openai; chat ?bots?; agentic; prompt (engineering|design|writing|engineer); natural language processing; computer vision; retrieval[-]augmented generation; vector databases?; transformer models?; mlops; (github|microsoft|m365|microsoft 365) copilot; copilot studio; google gemini; claude ai
Case-sensitive	A\.?I; LLMs?; GPT(-?\d(\.\d)?o?)?; NLP; GenAI; ML/AI; ML (?:models?|engineer\w*| pipelines?|platforms?|systems?|algorithms?|infrastructure|research\w*|scientists?|ops| solutions?|techniques?|methods?|frameworks?|applications?|features?|products?)
Near-miss (any case)	automat; model(s|ing|ling)?; agent(s|ic)?; copilot; prompt(s|ing|ed)?; bots?; algorithm; intelligen; predictive; data scien; claude; gemini; perplexity; midjourney; dall-?e; synthetic data; llm

Table B4: Working with AI as a share of AI duty lines (%)

Year	Keyword lines	Every duty line	Corrected for catch rates
2019	4.9	4.6	4.9 [2.9, 9.2]
2020	5.4	5.2	3.1 [2.3, 5.8]
2021	4.4	4.3	3.5 [2.2, 6.7]
2022	5.4	5.2	4.0 [2.8, 8.4]
2023	6.6	5.9	5.0 [3.6, 8.4]
2024	10.7	9.1	7.4 [6.3, 9.6]
2025	17.7	14.5	12.5 [11.5, 14.0]
2026	30.4	25.0	21.5 [20.4, 22.8]

Notes: Keyword lines: duty lines with an AI keyword, by Sonnet 5.5's side. Every duty line: Sonnet's labels for keyword lines and the classifier's for the rest. Corrected: keyword lines plus the AI lines without a keyword estimated from the random sample of keyword-free lines alone (Table B5; the text's estimates add the stratum of near-miss lines, Appendix B.3), with 95% intervals from the sample.

Table B5: Share of AI work caught by the keyword rules

Year	Building AI	Working with AI	Any AI work	Keyword AI lines
2019	39 [33, 46]	43 [20, 69]	42 [36, 48]	4,744
2020	37 [32, 43]	73 [32, 94]	42 [36, 48]	3,688
2021	39 [33, 45]	56 [26, 82]	44 [38, 49]	7,309
2022	50 [42, 58]	68 [28, 92]	51 [44, 59]	4,273
2023	47 [41, 54]	66 [35, 88]	49 [43, 56]	4,699
2024	50 [45, 56]	81 [59, 92]	56 [51, 61]	18,640
2025	55 [51, 59]	88 [78, 94]	62 [58, 65]	49,217
2026	51 [48, 54]	87 [82, 91]	62 [59, 64]	70,102

Notes: Percent of AI duty lines that contain an AI keyword, with 95% intervals from the binomial interval of the random sample's AI rate. Keyword AI lines: duty lines with an AI keyword that Sonnet labels AI work. Random sample: 18,750 keyword-free duty lines per year; the catch rates in the text add the stratum of near-miss lines (Appendix B.3).

B.4 A map of duty space

To show where duties sit and where they change, I draw 40,000 duty lines a year from 2019 to 2026 at random, embed their 287,234 distinct sentences with the same model, and place them in two dimensions with UMAP (cosine distance, 30 neighbours; McInnes et al., 2018). A map of all 10.1 million distinct sentences would be impractical to fit and unreadable as a scatter, and drawing the same number of lines each year gives every year equal weight. Nearby points describe similar work; the axes have no units. Figure B4 names 30 regions of the map (k-means clusters of the map, named by Sonnet 5.5 from random sentences in each, without dates). Figure B5 shows where the share of sampled duty lines grew and shrank between 2019–22 and 2025–26, with the occupation mix held fixed: each line of 2025–26 is weighted by its SOC major group's share of the 2019–22 lines over its share of the 2025–26 lines, so the map shows changes in what occupations ask for rather than in which occupations post. The labelled clusters are among 300 finer clusters, with changes of at least four standard errors and at least 15 employers on the side that is larger. One cluster of AI work gained 1.2 percentage points of the sampled lines, more than any other part of the map; recruiting and nursing-documentation duties lost most (0.22 and 0.21 points), while patient and family care coordination gained (0.16). Figure B6 places the AI duties of 2026 on the map: most sit at its top edge, next to data and engineering work, and some in sales, marketing, administrative

and patient-care regions.

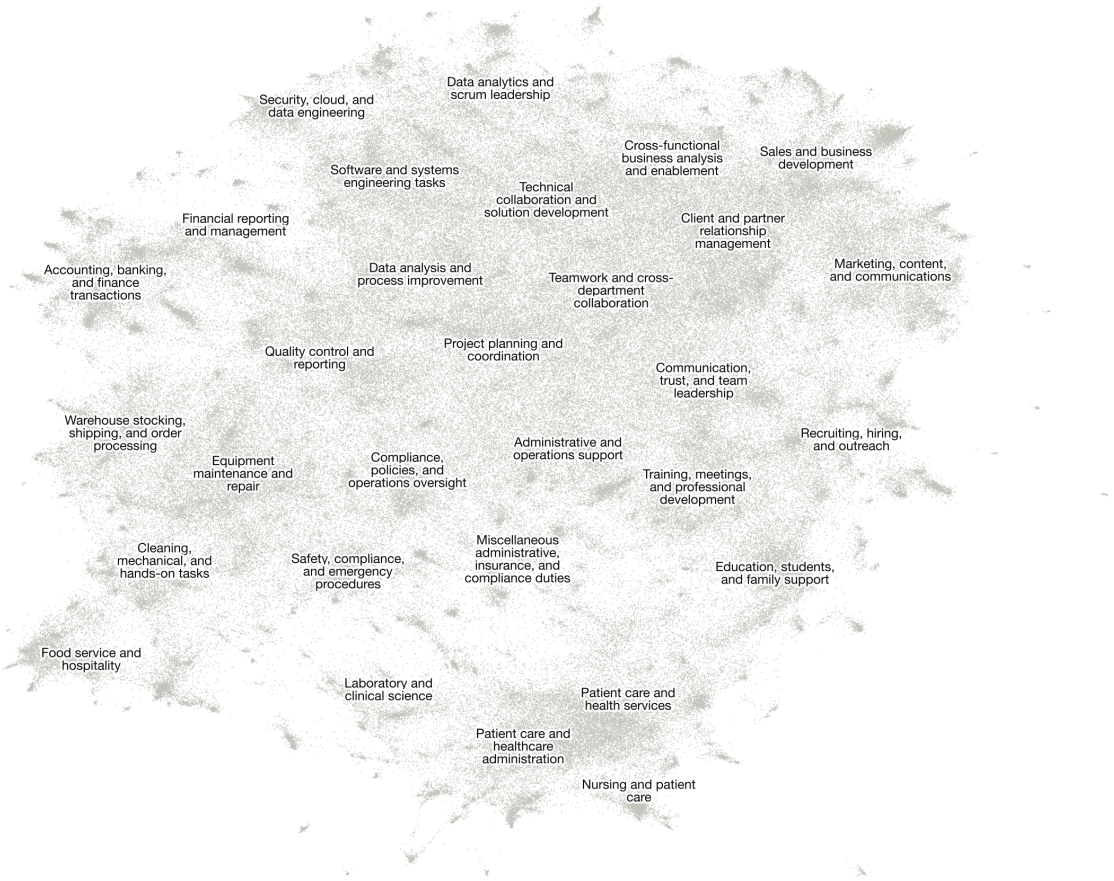


Figure B4: A map of duty space: 287,234 distinct duty sentences from a random sample of 320,000 duty lines (40,000 a year, 2019–2026), placed by meaning (sentence embeddings, UMAP). Region names by Sonnet 5.5 from random sentences in each of 30 regions.

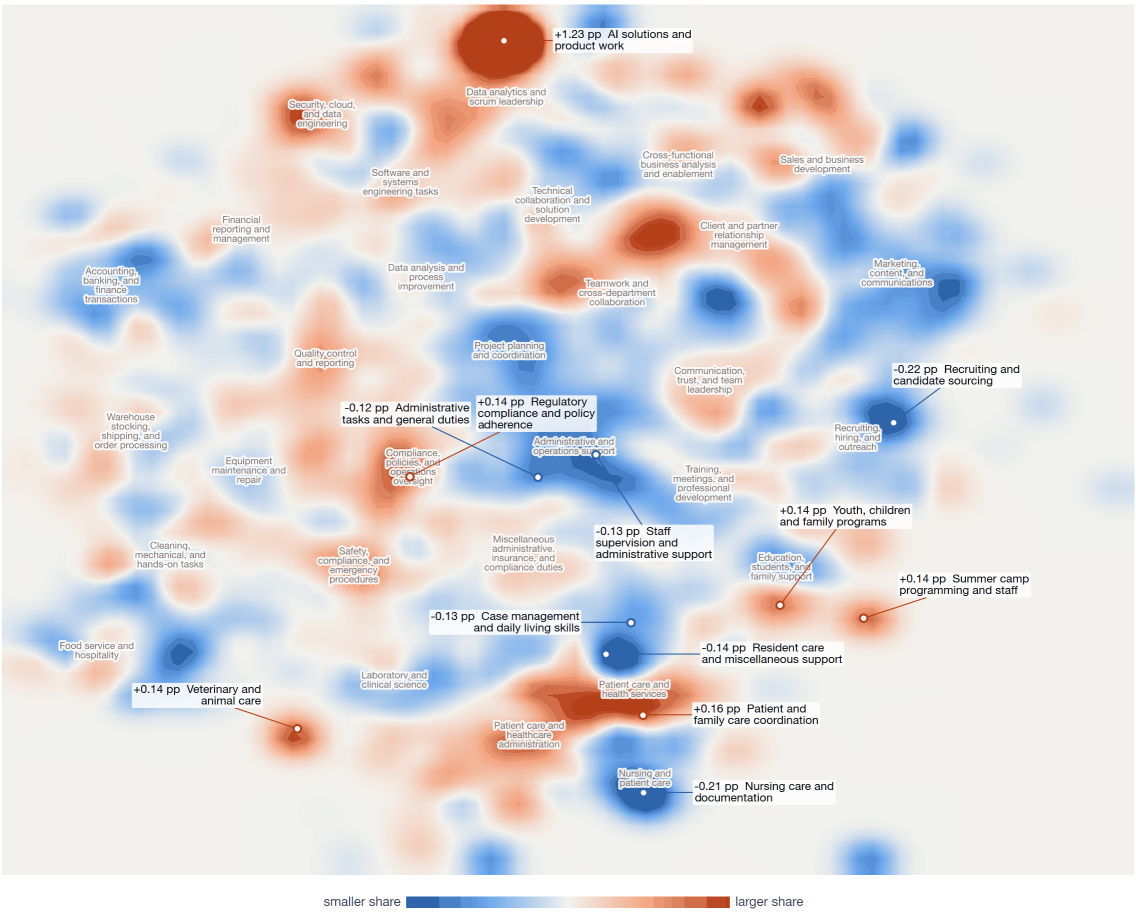


Figure B5: Where duties grew and shrank between 2019–22 and 2025–26, with the occupation mix held fixed: smoothed change in the share of sampled duty lines across the map of Figure B4. Labelled: the largest gains and losses among 300 finer clusters, with changes of at least four standard errors and at least 15 employers, in percentage points of all sampled lines; names by Sonnet 5.5.

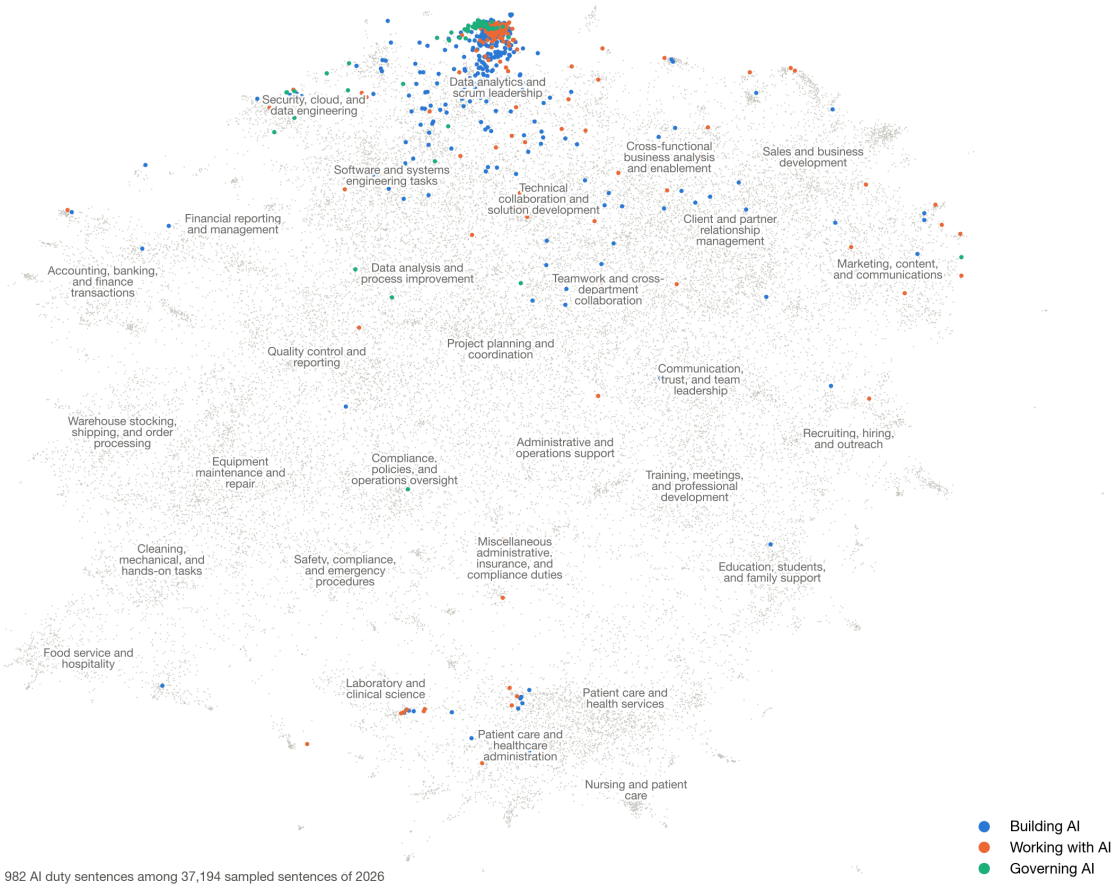


Figure B6: AI duties of 2026 on the map of Figure B4, by side (Sonnet 5.5 labels of sentences with an AI keyword). Grey: all sampled sentences of 2026.

B.5 Duty lines and O*NET tasks

Figure B7 shows, for the duty lines of 2019–26, the share whose closest task statement in the posting's own O*NET 27.1 list describes related work, and Figure B8 shows how the matching thresholds were set. Sonnet 5.5 graded 85 duty lines in each bin of 0.05 of similarity to their closest O*NET task, separately for the closest task among all occupations and among the posting's own occupation, with the prompt of Appendix B.6 (2: the same work; 1: related work; 0: unrelated). The thresholds where a logistic fit of the share graded at least related work crosses one half are 0.44 among all tasks (95% bootstrap interval 0.42–0.46) and 0.35 within the occupation (0.32–0.37). The share graded the same work crosses one half only at 0.73 and 0.71, at the top of the graded range, so the paper uses no threshold for the same work. Fitted on 400 pairs of new AI lines alone, the threshold within the occupation is 0.36 (0.33–0.38), close to the pooled one. For the five closest tasks, Sonnet graded 160 lines of each class whose closest task falls below the threshold, each against its five closest tasks in its occupation's list; at the standard of the same work, it finds such a task for 24% of these building-AI lines and 9% of the working-with-AI lines. The threshold counts every line above it as having a related task. Grading 300 such lines of each class directly, the closest task is related for 65–75% of them and one of the five closest for 92–97% (Opus: 86–93%); five tasks drawn at random from the same list are related for 67–77% of lines, so the lists' coverage of the new duties is well above chance, and the standard is loose. At the stricter standard of the same work, the five closest tasks of these lines describe 50% of continuing lines, 35% of new building-AI lines and 27% of new working-with-AI lines, against 7–15% for five random tasks. Integrating the fitted probability over each class's similarities instead of cutting at the threshold gives 73% of continuing lines, 50% of new building-AI lines, 53% of new working-with-AI lines and 58% of new non-AI lines with a related closest task, against 89, 49, 58 and 67% at the cutoff. Of the non-AI lines read as new tasks, 20% have no related task among the five closest (405 lines). O*NET revises occupations on a rolling cycle, and 27.1 still carries many lists written before 2019: where the list was last revised before 2019, the closest task is related for 40% of new building-AI lines and 49% of new working-with-AI lines, against 52% and 64% where it was revised in 2019 or later (continuing lines: 88% and 90%). O*NET 31.0, the release of August 2026, has 103 task statements that 27.1 lacks, 0 of them about AI, and matching to it changes none of the shares by more than a point. Table B6 lists the O*NET task statements that the new duty lines of 2025–26 in high-use occupations are most often matched to, by the number of employers.

Exposure across occupations. Across 213 detailed occupations, the share of duty lines read as new tasks has a rank correlation of 0.69 with the published exposure measure of Eloundou et al. (2024) and 0.78 with observed AI use in Massenkoff and McCrory (2026).

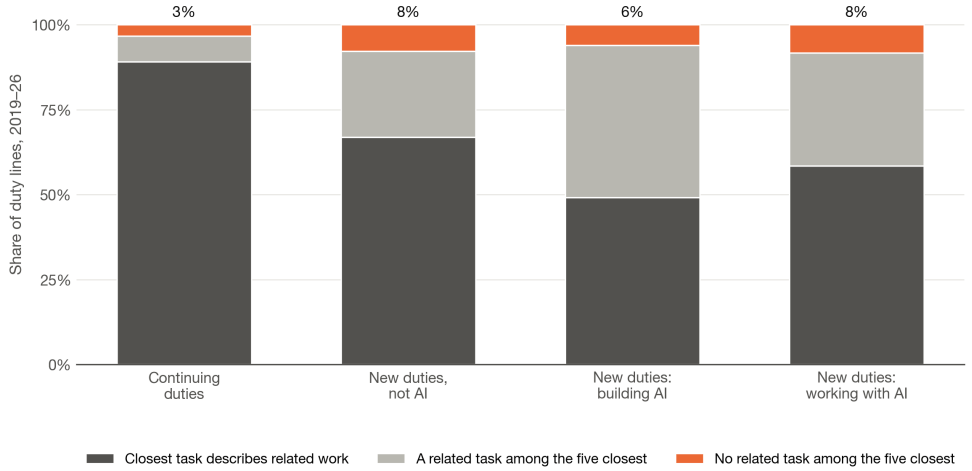


Figure B7: Duty lines of 2019–26 and the O*NET 27.1 task list of the posting's own occupation, by class of line: the closest task statement describes at least related work (dark; the threshold of Figure B8); otherwise, Sonnet 5.5 grades at least one of the five closest tasks as related work (light); or none of the five is related (orange, with its share). Random samples of continuing and new non-AI lines and every new AI line; the five closest tasks of 160 lines per class.

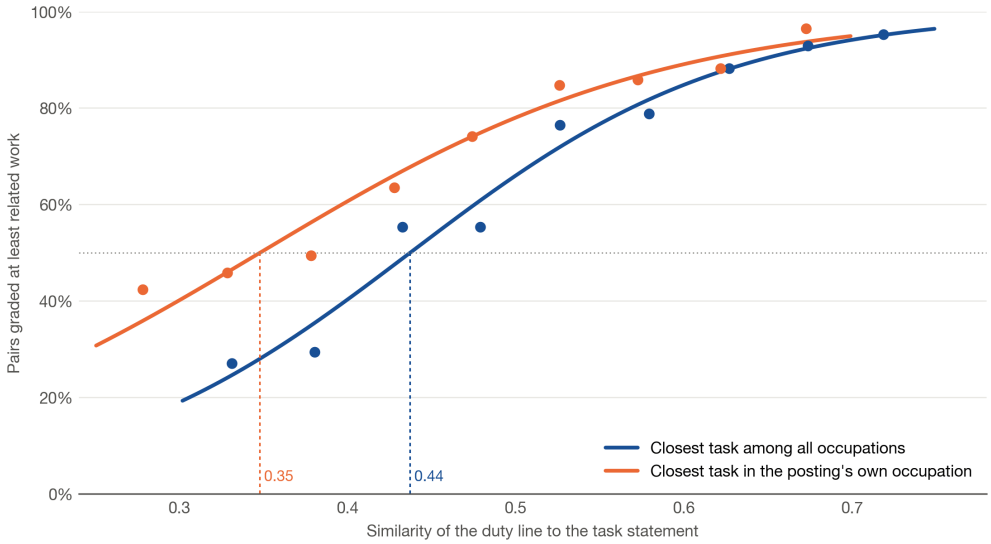


Figure B8: Setting the O*NET matching thresholds: share of duty-line and task pairs that Sonnet 5.5 grades at least related work, by similarity, with logistic fits, for the closest task among all occupations and in the posting's own occupation; 85 pairs per bin. Dotted lines: the thresholds, where the fits cross one half.

Table B6: O*NET tasks closest to new duty lines, 2025–26, occupations with high AI use

Closest O*NET 27.1 task	Employers	Lines
A. New AI duty lines		
Develop, test, or program new robots.	687	1,332
Create back-ups of robot programs or parameters.	404	680
Train others to install, use, or maintain robots.	388	642
Create business intelligence tools or systems, including design of related datab...	353	487
Collaborate with system architects, software architects, design analysts, and ot...	283	414
Collaborate with product development teams to research, model, validate, or impl...	260	404
Help clients improve decision making, abstract reasoning, memory, sequencing, co...	235	353
Develop or apply data mining and machine learning algorithms.	226	357
Research computerized automotive applications, such as telemetrics, intelligent ...	202	277
Develop defense plans or tactics, using intelligence and other information.	202	353
B. New duty lines not about AI		
Develop or implement marketing strategies.	530	696
Provide expertise to help financial institutions manage risks and maximize retur...	529	1,048
Collaborate with vendors to obtain or develop desired products.	508	881
Provide customers or installation teams guidelines for implementing secure syste...	482	917
Provide customers with ongoing technical support.	474	787
Collaborate with product development teams to research, model, validate, or impl...	451	729
Serve as liaisons between organizations, shareholders, and outside organizations...	429	601
Implement security practices to preserve assets, minimize liabilities, or ensure...	386	567
Develop or track quality metrics.	356	464
Provide training and support to system designers and users.	355	552

Notes: New duty lines (no close counterpart in 2019H1 postings) whose closest O*NET 27.1 task statement, among all occupations, describes at least related work by the threshold of Figure B8; ranked by distinct employers.

B.6 Language-model labels

Table B7 lists every labelling task. All labels come from Claude Sonnet 5.5, with Opus 5.5 as the second rater, except the occupation codes, which come from Opus 5.5 with Sonnet 5.5 as the second coder, and the checks that use a new prompt, which Opus 5.5 reads alone. The models are called through Anthropic's command-line client at medium reasoning effort, with structured output and every response cached, and they never see the date of the posting being labelled (the theme prompt marks example sentences as recent or older); Table B8 gives the model identifiers and the other specifications.

The new-line readings are made in batches of 40 lines that mix the AI and non-AI samples and every year and third. The batch matters: the same lines are read as new tasks more often when mixed than beside similar lines (66% against 47% for AI lines and 10.0% against 6.7% for non-AI lines, on 530 lines read both ways), and one line per call falls in between (60%). Read in mixed batches, the 14,646 lines of the two samples that are new under the threshold give new-task shares of 2.9% for non-AI lines (3.1% in one-class batches), 50% for building AI (31%) and 84% for working with AI (63%); the paper uses the mixed-batch readings, and Appendix D gives the headline on the one-class readings. Opus 5.5 reads a random fifth of the same batches (2,929 lines): it agrees with Sonnet on 84% of non-AI lines, 75% of building-AI lines and 87% of working-with-AI lines, with κ for the new-task class of 0.43, 0.66 and 0.64, and calls 1.9%, 45% and 78% of them new tasks. On non-AI lines the positive class is rare, so agreement on it is the least certain part of the reading: on 1,414 lines drawn to include every non-AI line that Sonnet called a new task, κ for that class is 0.46 (0.39–0.54) and Opus calls 0.78 as many new tasks as Sonnet; for AI lines κ is 0.60 (F1 0.80). Reading a line beside its five most similar 2019 sentences and the most similar one from postings of its own occupation group, instead of the single closest, makes Opus stricter rather than looser: it calls 42% of 500 AI lines new tasks, against 51% beside the single closest.

The 500 new non-AI lines of each year and third were drawn on the thirds of the title-based occupation codes (Appendix A.3); with the final codes each line moves to its posting's third, and the share of each reading in a third is the average over the sampling cells it draws from, weighted by their new lines, with cells of fewer than 20 lines (9 of 128, holding 1.4% of new non-AI lines) taking their sampling third's share. Opus also re-reads random samples of the remaining tasks: on the near-miss and keyword-free samples it agrees with Sonnet on whether a sentence is AI work and on its side in 98% and 100% of cases (κ 0.85 and 0.78); on the threshold reading in 77% (κ 0.65); on the O*NET grades in 79% (κ 0.66); and on the themes' categories in 91% (κ 0.88). Two variants of the prompts check the reading: the AI prompt without the job title and occupation group, and the new-line prompt with its four kinds defined in other words, which changes the share of new tasks from 4.0% to 5.5% for non-AI lines and from 50% to 55% for AI lines (κ 0.63–0.67). A separate reading by Opus of the lines read as new tasks asks two questions, whether the activity is new and whether the tool or object is new: of the new AI tasks of 2026, 11% (building AI) and 18% (working with AI) describe a new activity, such as directing agents or evaluating model output, and 82% and 77% a familiar activity done with or on AI, such as drafting or coding with an assistant. Both count as new tasks here, as new products count as new work in Autor et al. (2024); in the high-use third in 2026 they are 0.40 and 2.31 of the 3.5 new tasks per 100 duty lines. The system prompts follow; line breaks are added for printing.

Table B7: Language-model labelling tasks

Task	Unit	Design	Labelled	Model	Second rating
Side and activity of AI work	Duty sentence with an AI keyword	Census	134,656	Sonnet 5.5	Opus 5.5, random 2,000: side 90%
AI work without a keyword	Duty line without an AI keyword	Random, 18,750 lines a year	139,346	Sonnet 5.5	Opus 5.5, random 1,372: 100%
AI work, near-miss words	Duty line with a near-miss word	Random, 5,000 lines a year	35,166	Sonnet 5.5	Opus 5.5, random 1,000: 98%
What new non-AI lines are	New non-AI line and nearest 2019H1 duty	Stratified: year × third of AI use, 500 a cell[a]	12,386	Sonnet 5.5	Opus 5.5, random 2,457: 84%
What new AI lines are	New AI line and nearest 2019H1 duty	Stratified: side × period, 400 a cell	2,400	Sonnet 5.5	Opus 5.5, random 472: 81%
Checks of the reading	Line and nearest 2019H1 duty	COVID-19 lines; lines just above the threshold; known new work; reworded prompt	3,684	Sonnet 5.5	Opus 5.5, random 20%: 71–86%
AI work without context	Duty sentence with an AI keyword	Stratified: occupation group, 400 a cell	1,200	Sonnet 5.5	Same side as with context: 87%
O*NET grades for AI lines and five closest tasks	Duty line and O*NET task	Stratified by similarity; lines without a related closest task	4,400	Sonnet 5.5	Opus 5.5, random 20%: 85–88%
Novelty threshold	Duty line and nearest 2019H1 duty	Stratified: 100 per 0.05 of similarity	1,000	Sonnet 5.5	Opus 5.5, random 500: 77%
O*NET matching thresholds	Duty line and closest O*NET task	Stratified: 85 per 0.05 of similarity, two searches	1,530	Sonnet 5.5	Opus 5.5, random 500: 79%
Theme names and kinds	Rising theme of new lines	The 60 that rose most and the 40 largest not about AI	90	Sonnet 5.5	Opus 5.5, all: 91%
Region names	Region or cluster of the map	Census of the 30 regions and the labelled clusters	51	Sonnet 5.5	
Seniority	Job title	Half in proportion to postings, half uniform	30,000	Sonnet 5.5	Title rules: 87%
Industry	Firm	Census of firms with at least three postings	30,539	Sonnet 5.5	Second pass on low or medium confidence
Duty lines (text layer)[b]	Every sentence of a posting	100 postings a year and 200 with flat text	1,200	Sonnet 5.5	Retest of 100: 99%
Occupation codes[b]	Posting	Stratified by platform and year, plus AI and new titles	63,000	Opus 5.5	Sonnet 5.5, random 20%: detailed 73%
Technical and data tools	Duty line naming a tool, with the job title	Stratified: family × period × earlier use, 80 a cell	1,440	Opus 5.5	Also read the examples of Figure 8

Notes: Labelled: items labelled (sentences, lines, pairs, themes, titles, firms or postings). [a]Eight years (2019: second half) by the three thirds of observed AI use, and 250 a year in occupations without a measure. [b]From the dataset (Section 2).

Table B8: Models and specifications

Component	Specification
Language models	claude-sonnet-5-5 (reader) and claude-opus-5-5 (second reader; new prompts), through the Claude Code command line 2.1.283 in print mode with JSON-schema output, reasoning effort medium; calls made 2026-10-11 to 2026-10-17
Prompts	printed in Appendix B.6; every reading is released with its model and prompt
Sentence embedding	sentence-transformers/all-MiniLM-L6-v2, mean pooling, 128 tokens, L2-normalised; larger-model check: BAAI/bge-base-en-v1.5
Novelty screen	maximum cosine to the 679,314 distinct duty sentences of postings dated to 2019H1; threshold 0.66 (10% of 2019H2 lines new)
AI-activity classifier	logistic regression (stochastic gradient descent, log loss, alpha 1e-6) on word 1–2-gram TF-IDF (300,000 features) and the MiniLM embedding (weight 3); trained on all Sonnet-labelled sentences (keyword, near-miss and random samples); Sonnet labels override its predictions where they exist
Themes	spherical k-means on MiniLM vectors, 1,500 clusters, 20 iterations, seed 7, fitted on the distinct new sentences of 2025–26; every new sentence of 2019–26 assigned to its nearest centroid; rising: employer share in 2025–26 at least 3 times 2019–22 and at least 20 employers
O*NET matching	O*NET 27.1 task statements, MiniLM; related-work thresholds 0.348 (own occupation) and 0.437 (all), each fitted on 765 Sonnet-graded pairs of its own lookup

AI work: side and activity. Each call sends 60 items as "id | job title | occupation group | sentence".

```
You code duty sentences from US job postings for research on how AI changes work. Each item gives a
  job title, an occupation group and one duty sentence from that posting. Code every item.

ai: true if the sentence describes work that involves AI or machine-learning systems: generative AI,
  LLMs, chatbots, copilots, AI agents, machine-learning or deep-learning models, computer vision,
  speech or NLP systems, AI-driven automation. false for automation without AI (scripts, RPA,
  macros, workflow rules), statistics or analytics without ML, "models" meaning financial, data,
  business or 3D models, "agents" meaning people (insurance, customer service, real estate), and
  "AI" in other senses.

activity (if ai is true, the main one; if ai is false, "none"):
  build: research, design, develop, train, fine-tune or evaluate AI models, agents or AI-powered
    products and features that the organisation builds
  deploy: integrate AI into products, systems or infrastructure; MLOps; data pipelines for AI;
    operate, maintain or monitor AI systems in production
  sell_support: sell, market, consult on, implement for clients, or support AI products or services
  govern: AI policy, risk, compliance, ethics, security, legal or responsible-AI oversight
  use_tool: use AI tools to do the person's own work (drafting, coding with an assistant, analysis,
    research, design, answering customers)
  direct_review: direct, prompt or orchestrate AI agents or tools, or review, verify, correct or
    approve AI-generated output, as part of the person's own work
  adopt_enable: find AI use cases, lead or drive adoption of AI tools in a team or function,
    redesign workflows around AI, or train and coach colleagues to use AI
  other_ai: anything else involving AI (e.g. "stay current on AI trends")

gen (if ai is true): "generative" if it clearly involves generative AI (LLMs, chatbots, copilots,
  generative agents, text, image or code generation); "other_ml" if it is clearly other machine
  learning; "unclear" otherwise. If ai is false: "na".
```

New lines and the novelty threshold. Each call sends 40 items as "id | job title | recent sentence | most similar 2019 sentence".

```
You compare a duty sentence from a recent US job posting with the most similar duty sentence found
  in US job postings from early 2019, to judge whether the recent sentence describes new work. For
  each item give:
kind:
  new_task: the recent sentence describes an activity, responsibility or object of work that the
    2019 sentence does not, and that would not have been common in 2019 job postings for this kind
    of job
  variant: the same kind of activity as the 2019 sentence, with a different object, tool, setting or
    level of detail that is not new in itself
  reworded: the same activity as the 2019 sentence, in different words
  not_a_duty: the recent sentence is not a description of work (company, benefits, role context,
    requirements, or a fragment)
source (only for new_task; otherwise "na"): what makes it new: technology (non-AI tools, platforms,
  data systems), regulation_compliance, health_safety, product_market (new products, services,
  markets or business models), organisation_people (e.g. diversity, remote or hybrid work,
  wellbeing, culture), sustainability, other
```

Technical and data tools. Each call sends 40 items as "id | job title | tool family | duty sentence".

```
You read duty sentences from US job postings that name a technical or data tool, each with the job
  title and the family of tools it was matched to. For each item give:
uses_tool: "yes" if the sentence asks the person in this job to use, build or work with a tool of
  that family (for example, building dashboards in Tableau, writing SQL queries, automating a
  process with Power Automate); "no" if the tool is named for another reason (the employer's own
  product, what another team uses, a qualification, or a word that is not the tool, such as a
```

```
  therapy called DBT or a dashboard of a vehicle)
activity: what the person does with the tool: dashboards_reports, data_analysis, code_scripts,
  automate_workflows, manage_data, or other
ai: "yes" if the sentence also involves AI tools or models, otherwise "no"
```

O*NET matching thresholds. Each call sends 20 duties, each with its job title and one candidate task statement.

```
You judge how well standard task descriptions from the O*NET database describe a duty from a US job
  posting. For each candidate, give a grade:
2 = it describes the same work as the duty (wording may differ; small differences in scope are fine)
1 = related work (same area or a closely connected step), but not the same work
0 = unrelated, or only superficially similar
Judge the work itself, using the job title for context. Grade every candidate independently.
```

Themes. Each call sends up to eight recent and four older example sentences per theme.

```
You classify recurring duties in US job postings. Each group is a set of similar duty sentences from
  different postings: "recent" sentences, and the "older" sentences closest to the same theme. For
  each group give a short name (at most 8 words, sentence case, describing the recent work) and one
  category for the recent sentences:
ai_building: building, deploying, integrating, selling or supporting AI systems (models, agents, LLM
  products)
ai_using: using, prompting, directing, reviewing or driving the adoption of AI tools in the worker's
  own work
ai_other: governing AI, or other work about AI
new_task: genuinely new work that is not about AI and would not have been common in 2019 postings
  for such jobs
variant: a familiar kind of work with a new object, tool, platform or setting that is not new in
  itself
reworded: familiar work in new wording, or a shared template
not_work: culture, values, benefits, notices or generic statements rather than work
source (only for new_task; otherwise "na"): technology (non-AI), regulation_compliance,
  health_safety, product_market, organisation_people, sustainability, other
```

B.7 Seniority, industry and firms

Seniority levels are intern, junior, mid, senior, staff or principal, manager, director and executive. Title rules alone agree with Sonnet on 87% of 1,000 titles; the title model, trained on 30,000 Sonnet-labelled titles (half drawn in proportion to postings, half drawn uniformly from the remaining titles), agrees on 88% in cross-validation. Industry codes were checked and recoded where Sonnet's confidence was low or medium, using the employer's company-description sentences; 77% of firms (90% of postings) are coded with high confidence. Employer accounts are linked across platforms when a normalised name is unique to one account on each of at least two platforms; the links rest on the name alone, 5,040 accounts become 2,414 firms, and every other firm is a single account.

C Further results

Table C1 collects the headline comparisons under other choices. Two further checks bear on what the new AI lines are. They are not boilerplate copied across employers: 2% of the working-with-AI lines of 2025–26 recur word for word at a second firm, against 10% of continuing lines, and among

Table C1: Robustness of the headline numbers

	New AI, high	New AI, low	High/low	Δ new non-AI	Working with AI (%)	Postings (%)	Exposure design
A. Weights, platforms, threshold and period							
Baseline	3.73	0.26	14	+2.6	25.0	5.9	1.65 (0.42)
OEWS employment weights	1.83	0.08	22	+1.8	29.9	2.7	0.79 (0.32)
CPS new-hire weights	1.69	0.08	21	+1.7	30.8	2.6	0.60 (0.26)
Without Workday and Lever	3.76	0.27	14	+3.2	25.3	6.1	1.72 (0.43)
Fixed platform mix	2.84	0.17	17	+1.5	26.7	4.6	1.60 (0.42)
Novelty threshold −0.02 (6% of 2019H2 new)	2.80	0.20	14	+1.9	25.0	5.9	
Novelty threshold +0.02 (14% of 2019H2 new)	4.66	0.32	14	+3.3	25.0	5.9	
Without 2026H2	3.71	0.27	14	+2.4	23.7	5.5	1.14 (0.29)
B. Reference, embedding model, sample and occupation codes							
Novelty threshold: 5% of 2019H2 new	2.25	0.16	14				
Novelty threshold: 20% of 2019H2 new	5.42	0.36	15				
Reference without January 2019	3.27	0.22	15				
Larger embedding model (sample)	3.10	0.26	12				
Firms posting in 2019H1 and 2025–26	3.61	0.29	13				
Without computer occupations	1.59	0.26	6				1.60 (0.54)
Major group × half-year effects							0.81 (0.42)
Confident occupation codes	4.04	0.24	17				
Codes with probability ≥ 0.75	4.23	0.20	22				
AI use over the three likeliest codes	3.60	0.16	23				
Title-based occupation codes	3.71	0.35	10				

Notes: New AI, high and low: new AI duty lines per 100 duty lines in 2026, before reading, thirds of occupations with the most and least observed AI use; High/low: their ratio. Δ new non-AI: change in new non-AI lines per 100 duty lines, high-use occupations, 2022 to 2026 (raw, not read). Working with AI: share of AI duty lines, 2026, every duty line counted (25%; the text's 22% also allows for the keywords' catch rates, Table B4). Postings: share of postings with a working-with-AI duty, 2026. Exposure design: post (2024H1 on) minus pre (to 2022H2) coefficient for postings with a working-with-AI duty, per standard deviation of AI use (standard error), with occupations weighted by their 2019–22 postings or, in the OEWS and CPS rows, by their employment (OEWS, May 2025) and their new job starts (CPS, 2019–22, by major group, split by employment). The novelty threshold does not affect the working-with-AI share or the postings column, which count all AI lines. Panel B: each threshold makes the stated share of 2019H2 lines new; the larger embedding model (BAAI bge-base) reads 20,000 lines per half-year with its own threshold; firms: linked employer accounts posting in both periods; without computer occupations: SOC 15 dropped, thirds kept; the last three rows cut the thirds on codes with a probability of at least 0.75 (97% agreement with Opus 5.5), on each posting's expected AI use over its three likeliest codes, and on the title-based codes (Appendix A.3).

distinct sentences, allowing for paraphrase, 2.1% of working-with-AI lines and 1.4% of building-AI lines have a near-duplicate (cosine similarity of at least 0.90) at another firm, against 2.1% of continuing lines and 1.0% of new non-AI lines, on 20,000 distinct sentences of each kind. And they do not come from new employers: in the firms that posted both in early 2019 and in 2025–26, new AI lines are 1.6 in every 100 duty lines in 2026, against 1.7 in all postings, with 3.6 in the high-use third and 0.29 in the low-use third.

Two other counts of new work are broader. Postings in titles that at least three employers used in a year and that no posting of 2019–22 used, in any spelling, rose from 0.1% of postings in 2023 to 0.7% in 2026, and to 1.2% in the third of occupations with the most AI use; 46% of them have an AI duty. Duty lines new to a firm and occupation, the analogue in text of the new skills counted by Chen et al. (2024), are 55–59% of the duty lines of each year from 2023; 77–83% of them are close to a duty of early 2019, and Sonnet reads 3% of the rest that are not about AI as new tasks.

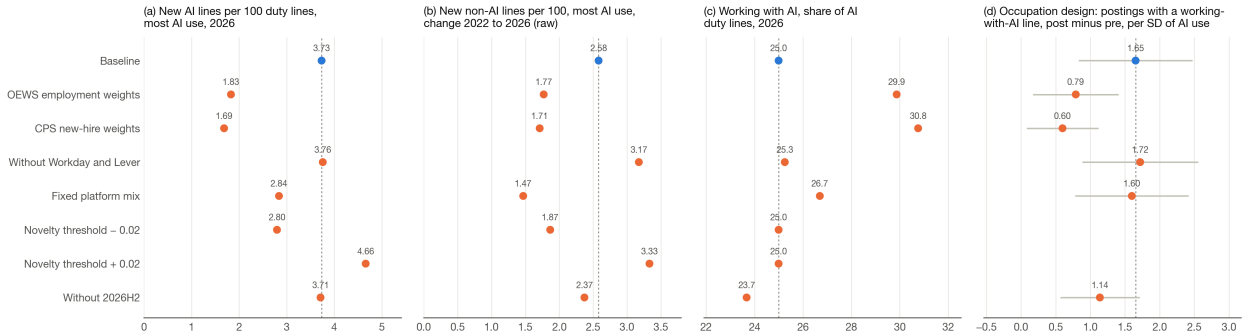


Figure C1: Four headline numbers of Panel A of Table C1 under each variant: (a) new AI lines in the high-use third, (b) the change in new non-AI lines, (c) working with AI as a share of AI duty lines (every line counted) and (d) the exposure design. Blue: baseline.

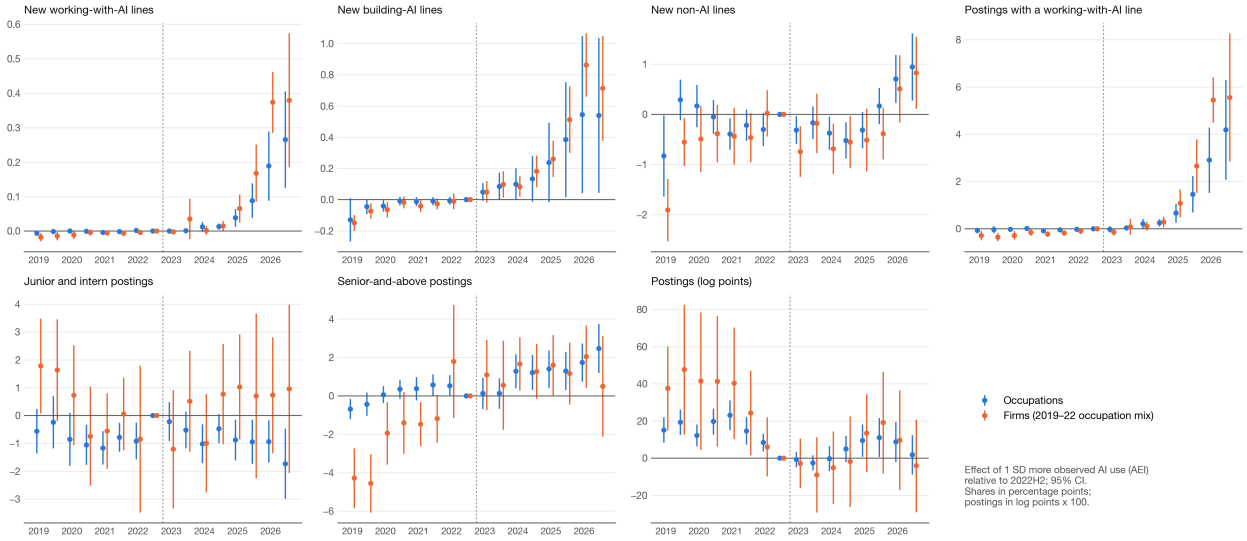


Figure C2: Occupation-level design for every outcome, and a firm-level version in which a firm's AI use is that of its 2019–22 occupation mix (230 occupations; 1,756 firms). Senior and above includes managers, directors and executives.

Table C2: The activities of AI duty lines by year (%)

Year	Build	Deploy	Sell, support	Use tools	Direct, review	Drive adoption	Govern	Lines
2019	72.5	15.7	5.5	3.4	0.4	1.8	0.6	4,186
2020	71.1	16.7	5.2	4.0	0.4	2.0	0.7	3,142
2021	63.4	22.5	7.4	3.2	0.3	1.8	1.4	6,095
2022	64.4	20.0	7.3	3.6	0.3	2.1	2.3	3,835
2023	63.8	16.6	9.7	3.5	0.9	3.1	2.4	4,195
2024	55.3	16.7	11.4	4.5	2.8	4.7	4.6	16,718
2025	46.0	17.5	12.0	8.4	0.9	10.4	4.9	44,391
2026	32.9	14.3	12.4	14.1	1.9	17.4	7.0	63,951

Notes: Sonnet 5.5 labels of every duty sentence with an AI keyword, as shares of the AI duty lines with an activity. Lines: AI duty lines with an activity. 2026: January to September.

C.1 Firm event studies

If working with AI spreads within firms, it should appear in a firm's other jobs once the firm starts building AI. I compare firms whose first posting for AI-building work falls in 2023–25 with firms that start later, before those firms' own first such posting, and do the same around the first posting for working with AI, an event close to the adoption measure of Hosseini and Lichtinger (2026). I estimate the event study on a balanced panel of firms and quarters, with quarters without postings filled in, with the estimators of Callaway and Sant'Anna (2021), with not-yet-treated firms as controls and cohorts observed for six quarters after the event, and of Sun and Abraham (2021), with firms that start in 2026 as the comparison cohort. Figure C3 and Table C3 show the results for 458 firms that first post AI-building work in 2023–25 (604 firms in all) and 632 that first post working-with-AI work (1,033); the balanced Callaway–Sant'Anna estimates use the cohorts of 2023Q1–2024Q2, which are observed for six quarters after their event.

Neither robust estimator finds a detectable change. Working-with-AI duties in the firms' non-technical postings move by +0.4 points with Callaway and Sant'Anna's estimator and +0.8 with Sun and Abraham's (standard errors 0.3 and 0.5), against 1.2 in a stacked design on the unbalanced panel with later adopters as controls (Cengiz et al., 2019), with no difference before the event. The junior share of their other postings moves by −1.0 and −2.4 points (standard errors 1.5 and 1.6) on a base of 17.8%. Around the first working-with-AI posting, the junior share moves by −0.4 points with Sun and Abraham's estimator (standard error 1.1), while Callaway and Sant'Anna's balanced estimate is +0.3; Hosseini and Lichtinger (2026) measure junior employment from résumés, and these data measure the composition of postings. The design has a limit that no estimator removes: a firm's first such posting is more likely to be seen in a quarter when the archive captures many of its postings, so the event coincides with a burst of postings and posting volumes trend before it. Dating the event by the stated posting date, as here, cuts that burst from 59 log points, when events were dated by the first capture, to 26, and leaves the estimates where they were. Restricting the panel to firms with at least three or five captured postings in every quarter leaves 37 and 17 firms, too few to estimate anything, and conditioning on the firm's log captured postings changes nothing. Share outcomes are undefined in quarters without postings: the balanced estimates fill those quarters with zero postings and a share of zero, and estimates on the observed firm-quarters alone give the same answer (working with AI in non-technical postings +0.9 points, standard error 0.5; the junior share −1.3, 1.4).

Table C3: Firm event studies: post minus pre

Outcome	Stacked	Callaway–Sant'Anna	Pre-trend p	Sun–Abraham	Pre-trend p
A. Around the first AI-building posting					
Working with AI, non-tech postings	1.2 (0.4)	0.4 (0.3)	0.44	0.8 (0.5)	0.46
Intern or junior share	−0.9 (1.1)	−1.0 (1.5)	0.80	−2.4 (1.6)	0.94
Senior or staff share	−0.8 (1.2)	1.2 (1.4)	0.89	−0.1 (1.5)	0.75
Computer occupations share	−0.5 (1.0)	−1.9 (1.6)	0.63	−1.2 (1.4)	0.55
Postings (log points)	7.7 (5.7)	−3.2 (7.6)	0.00	21.9 (7.3)	0.00
B. Around the first working-with-AI posting					
Intern or junior share	−0.2 (0.8)	0.3 (1.0)	0.43	−0.4 (1.1)	0.78
Senior or staff share	0.3 (1.0)	−0.7 (1.4)	0.05	−0.5 (1.3)	0.98
Computer occupations share	0.6 (0.9)	0.3 (1.3)	0.80	−0.2 (1.2)	0.92
Postings (log points)	6.0 (4.2)	−15.4 (7.7)	0.00	4.4 (6.2)	0.00

Notes: Mean of the effects in quarters 0 to 6 minus the mean in quarters −8 to −2, in percentage points (postings: log points × 100), with standard errors. Stacked: stacked design with later adopters as controls (Cengiz et al., 2019), unbalanced panel. Pre-trend p: Wald test that the effects in quarters −8 to −2 are zero.

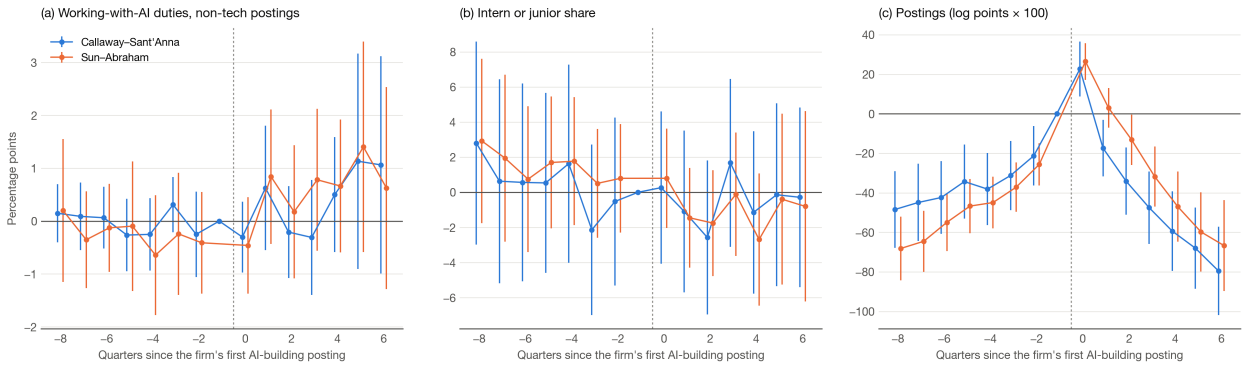


Figure C3: Firm event study around the first posting for AI-building work: coefficients by quarter relative to the event, with 95% intervals. Blue: Callaway and Sant'Anna's estimator with not-yet-treated controls, universal base period and cohorts observed for quarters 0 to 6. Orange: Sun and Abraham's estimator with firms that start in 2026 as the comparison cohort. Balanced panel of firms by quarter, 2019–2025, standard errors clustered by firm. Outcomes: (a) the share of non-technical postings without an AI-building duty that have a working-with-AI duty; among postings without AI duties, (b) the intern or junior share and (c) postings (log points × 100). Table C3 also reports the stacked design and the first working-with-AI posting.

Table C4: Posted-pay premia by occupation group

Fixed effects	Building AI	Working with AI	Postings
All postings			
Occupation × state × year	0.170 (0.012)	0.054 (0.008)	428,836
+ seniority	0.155 (0.012)	0.026 (0.007)	428,836
Firm × occupation × year	0.068 (0.004)	0.022 (0.008)	428,836
Firm × title × year	0.009 (0.007)	0.022 (0.014)	428,836
Computer and mathematical			
Occupation × state × year	0.142 (0.012)	−0.009 (0.011)	58,230
Firm × occupation × year	0.052 (0.005)	0.007 (0.009)	58,230
Management and business			
Occupation × state × year	0.203 (0.013)	0.083 (0.007)	120,551
Firm × occupation × year	0.093 (0.006)	0.029 (0.008)	120,551
All other occupations			
Occupation × state × year	0.199 (0.025)	0.074 (0.020)	250,055
Firm × occupation × year	0.078 (0.015)	0.032 (0.033)	250,055

Notes: Log annual posted pay, 2023–26, on indicators for a building-AI (build) and a working-with-AI (use) duty, with fixed effects for the cells in the first column. Pay is the midpoint of the posted range, annualised. Standard errors clustered by firm in parentheses. Postings: postings in the regression sample. Without year effects, the firm-and-title premia are 0.031 and 0.033; controlling for state laws that require pay ranges in postings changes no estimate by more than 0.01.

Table C5: The 30 largest rising themes of new duty lines

Theme (named by Sonnet)	Judged as	Employers 2019–22 (%)	2025–26 (%)	AI lines (%)
Integrate LLMs and AI into applications	Building AI	0.01	1.09	99
Build and operate AI platform infrastructure	Building AI	0.02	1.07	97
Leverage AI tools to work faster	Working with AI	0.01	1.00	99
Design and build AI-native workflows	Working with AI	0.01	0.96	98
Develop AI-driven product features and integrations	Building AI	0.07	0.95	96
Create and codify repeatable playbooks	Re-worded	0.22	1.06	2
Build agent platforms and tool layers	Building AI	0.03	0.87	81
Champion and drive team AI adoption	Working with AI	0.04	0.87	96
Using and adopting AI in workflows	Working with AI	0.01	0.83	99
Designing user interfaces for AI products	Building AI	0.04	0.85	97
Staying current on emerging AI capabilities	Governing/other AI	0.04	0.83	97
Building agentic AI workflows	Building AI	0.00	0.78	96
Positioning and marketing AI products	Building AI	0.03	0.80	89
Building AI agent systems and infrastructure	Building AI	0.02	0.76	100
Using AI-assisted development tools for coding	Working with AI	0.01	0.75	96
Designing and owning reliable automated workflows	Variant	0.33	1.05	5
Build and integrate LLM-based applications	Building AI	0.02	0.74	97
Ensure responsible, compliant AI governance and ethics	Governing/other AI	0.02	0.74	92
Build, launch, and sell agentic AI systems	Building AI	0.01	0.72	100
Build AI-ready data infrastructure and platforms	Building AI	0.10	0.82	93
Implement, evaluate, and improve LLM systems	Building AI	0.01	0.72	99
Lead and execute enterprise AI strategy	Governing/other AI	0.02	0.72	95
Generate and own sales pipeline	Re-worded	0.28	0.97	2
Sell AI and manage AI investments	Building AI	0.04	0.72	89
Build and operate agentic AI platforms	Building AI	0.01	0.69	98
Build backend services for AI applications	Building AI	0.03	0.70	99
Engineer AI and agentic solutions	Building AI	0.08	0.75	90
Use AI coding assistants in development	Working with AI	0.01	0.68	97
Establish governance frameworks and cadences	Re-worded	0.26	0.93	3
Use AI tools to accelerate routine work	Working with AI	0.02	0.68	96

Notes: Themes of the new duty lines of 2025–26 that rose at least threefold in the share of employers using them, ranked by the increase. Names and judgements by Sonnet 5.5. AI lines: share of the theme's lines labelled AI work. The themes are built on postings with confident title-based occupation codes (Appendix A.3) and dated by first capture; they are descriptive.

Table C6: The named rising themes that are not AI work

Theme (named by Sonnet)	Judged as	Employers 2019–22 (%)	2025–26 (%)	AI lines (%)
Create and codify repeatable playbooks	Re-worded	0.22	1.06	2
Designing and owning reliable automated workflows	Variant	0.33	1.05	5
Generate and own sales pipeline	Re-worded	0.28	0.97	2
Establish governance frameworks and cadences	Re-worded	0.26	0.93	3
Represent company at industry events	Re-worded	0.29	0.92	0
Develop market narrative and storytelling	Re-worded	0.28	0.87	2
Ship fast and iterate on product	Re-worded	0.13	0.69	3
Conduct customer and technical discovery	Re-worded	0.15	0.68	8
AI developer community advocacy and engagement	Variant	0.07	0.59	70
Build automated security and compliance guardrails	New task	0.06	0.58	17
Drive engineering improvements and initiatives	Re-worded	0.23	0.74	2
Own platform reliability, observability and incident practices	Re-worded	0.16	0.66	3
Build and own observability platforms	Variant	0.05	0.54	11
Make build-versus-buy and tradeoff decisions	Re-worded	0.16	0.64	2
Support go-to-market strategy and execution	Re-worded	0.15	0.63	3
Lead engineering analysis and integration	Re-worded	0.22	0.69	1
Go-to-market recruiting and GTM support	Variant	0.11	0.58	4
Executive engagement and communication	Re-worded	0.19	0.65	1
Surface insights and design across surfaces	Re-worded	0.09	0.55	4
Craft clear narratives for decision-makers	Re-worded	0.21	0.67	2
Drive readiness across operations and launches	Re-worded	0.11	0.55	2
Build trust and safety technology	Variant	0.15	0.58	4
Own systems and technical architecture	Re-worded	0.15	0.58	7
Build scalable co-selling and enterprise sales motions	Re-worded	0.05	0.47	2
End-to-end ownership of systems and outcomes	Not work	0.15	0.56	1
Unblock customers, teams, and deployments	Re-worded	0.18	0.58	1
Healthcare domain insight and clinical communication	Variant	0.13	0.53	4
Identify and reduce friction points	Re-worded	0.05	0.44	3
Identify and mitigate risks early	Re-worded	0.16	0.53	2
Partner with RevOps on tracking and reporting	Re-worded	0.07	0.44	1
Design mechanical systems for industrial robots	Variant	0.12	0.48	3
Optimize latency and throughput of real-time systems	Re-worded	0.10	0.46	13
Builders who shape and ship product	Not work	0.16	0.51	0
Define and refine ideal customer profile	Re-worded	0.08	0.43	1
Develop and maintain signals and detection tooling	Variant	0.15	0.51	6
Flag risks and blockers early	Re-worded	0.13	0.48	0
Build and support go-to-market motions	Re-worded	0.03	0.38	4
Improve win rates and balance quick wins	Re-worded	0.16	0.51	1
Drive cross-functional and systems alignment	Re-worded	0.15	0.49	2
Ensuring clean handoffs between teams	Re-worded	0.12	0.46	1

Notes: The 40 rising themes, of the 90 that Sonnet 5.5 named, that Sonnet 5.5 judges not to be AI work, ranked by the increase in the share of employers using them. Columns as in Table C5.

Table C7: The 20 largest new and rising job titles

Title	Kind	Employers, 2019–22	2024–26	AI duty (%)	New lines (%)	Pay ($k)
Forward deployed engineer	rising	9	247	61	64	175
AI engineer	rising	10	168	95	70	176
GTM engineer	new	0	118	78	93	162
Senior AI engineer	rising	7	103	98	87	191
Staff product manager	rising	13	94	38	42	215
Applied AI engineer	rising	1	78	100	87	212
Founding account executive	rising	2	61	42	65	195
Senior product engineer	rising	11	59	31	57	192
AI/ML engineer	rising	3	49	100	61	162
Senior design engineer	rising	8	48	17	68	175
Deployment strategist	rising	4	44	33	71	160
GTM recruiter	rising	6	41	18	25	150
Member of technical staff	rising	3	39	74	75	254
Principal product marketing manager	rising	5	37	38	37	181
Senior AI/ML engineer	rising	2	35	98	64	182
Staff electrical engineer	rising	4	34	2	41	186
Founding product designer	rising	6	33	53	29	180
HVAC maintenance technician	rising	4	31	0	31	73
AI product manager	rising	1	31	100	59	155
Plumbing service technician	rising	2	30	0	40	85

Notes: Normalised job titles with at least 10 employers in 2024–26 and none in 2019–22, in any spelling (new), or at least 20 employers in 2024–26 and five times their 2019–22 count (rising), ranked by employers in 2024–26; postings for talent pools and general applications excluded. AI duty: share of the title's postings with an AI duty. New lines: share of its 2024–26 duty lines with no close counterpart among the duties of its three nearest established titles. Pay: median posted annual pay.

D Checks

This appendix collects the checks of the headline measure, genuinely new work per 100 duty lines (Figure 1), under other choices of frame, dates, readings and screens (Table D1), and the checks of the other results. The paper's basis is postings dated by their stated date and readings in mixed batches (Sections 2 and 3.3); the frame variants use the first-capture frame and readings in batches of one class, each against its own baseline, and the table says so.

Table D1: Genuinely new work per 100 duty lines under other choices

	Most AI use, 2022	Most AI use, 2026	Least AI use, 2026	All occupations, 2026
Paper (stated dates, mixed-batch readings)				
Paper	0.74	3.48	0.33	1.72
Readings in batches of one class and cell	0.77	2.84	0.35	1.46
Opus 5.5's readings of a fifth of the batches	0.52	3.08	0.55	1.54
AI shares measured for 2019–22, the low third and governing AI	0.61	3.34	0.26	1.63
Composition (paper's basis)				
Firms posting in every year 2022–26	0.77	3.70	0.33	1.81
Inside firm × occupation cells present in both years	0.79	3.87		1.67
Without the 1% of firms with most postings	0.74	3.48		1.85
Occupations (paper's basis)				
Thirds on title-based occupation codes	0.77	3.28	0.38	1.70
Thirds of Eloundou et al. exposure	0.67	2.15	0.13	1.70
Screen (paper's basis; threshold rebuilt)				
Threshold: 5% of 2019H2 new ($\tau = 0.627$)	0.50	2.12	0.21	1.08
Threshold: 7.5% ($\tau = 0.646$)	0.63	2.82	0.29	1.41
Threshold: 12.5% ($\tau = 0.672$)	0.91	4.03	0.38	1.99
Threshold: 15% ($\tau = 0.682$)	1.06	4.48	0.43	2.23
Threshold: 20% ($\tau = 0.700$)	1.42	5.33	0.52	2.67
Reference without the January 2019 crawl ($\tau = 0.652$)	0.75	3.24	0.34	1.63
Frame and dates (first-capture frame, one-class readings)				
First-capture frame, one-class readings	0.74	2.77	0.37	1.42
Without postings with an unresolved location	0.73	2.76	0.37	1.41
Each posting in every half-year it stayed up	0.75	2.64	0.35	1.35
Dated by the stated posting date (within a year)	0.75	2.82	0.38	1.45
Dated by the stated posting date (within ten years)	0.77	2.88	0.38	1.50
Bulk-poster cutoff prorated for 2026	0.74	2.77	0.37	1.42
Without Workday (flat text)	0.72	2.80	0.37	1.45
Without Workday and Dayforce	0.72	2.85	0.38	1.48

Notes: Genuinely new work per 100 duty lines (Figure 1) in the thirds of occupations with the most and least observed AI use, and in all occupations. Each block states its basis: the Paper row is the basis of the Composition, Occupations and Screen blocks, and the first row of the last block is the basis of the frame variants.

Intervals and what counts. The 95% intervals of the headline, which cover the sampling of employers and of the read lines, are 0.51–0.98 for the high-use third in 2022 and 3.05–3.90 in 2026, or 2.42–3.90 allowing also for the second reader (Readings, below). In the high-use third the figure was 2.3 in 2025. In 2026, 24% of the postings of the high-use third have at least one duty read as a new task, against 7% in 2022 and 3% of the postings of the low-use third in 2026. Sonnet reads 42% of the new building-AI lines of 2026 as new tasks and 74% of the working-with-AI lines; if every new AI line counted as new work, the 2026 figure for the high-use third would be 4.7 rather than 3.5. Of the AI lines read as new tasks in 2026, 11% (building AI) and 18% (working with AI) describe a new activity and 82% and 77% a familiar activity done with or on AI (Appendix B.6).

Frame and dates. Dropping the 28,835 postings whose location could not be resolved, counting each advert in every half-year between its first and last capture (so that an advert that stays up, or is reposted unchanged, counts in later years too), dating postings by their stated date within ten years of capture rather than one, and scaling the bulk-poster cutoff to the nine months of 2026 each move the headline by little more than a tenth of a point, as does dropping Workday, whose pages

carry text only from 2022, or Workday and Dayforce, the platforms with flat text (Table D1). Of the postings of 2019–26, 88% carry a stated date within the year before their first capture and are dated by it; the first capture comes a median of 37 to 67 days after the stated date, with a 90th percentile of 154 to 475 days and a 95th percentile of up to 731, depending on the year.

Composition. In the 2,520 firms that posted in every year 2022–26, which hold 40% of 2026 postings, genuinely new work in the high-use third goes from 0.77 to 3.70 per 100 duty lines (all postings: 0.74 to 3.48), and working with AI is 25% of AI lines in 2026 (every line counted; all postings: 25%). Decomposing the change over firm-by-occupation cells, cells present in both years hold 22% of 2026 lines, since the archive's thin sample makes cells come and go, and inside them the rate goes from 0.79 to 3.87; cells entering in 2026 are at 3.36. The rise happens inside firms and occupations at the rate of the whole. The top 1% of firms by 2026 postings (145 firms) hold 24% of 2026 postings and 15% of 2026 AI lines, and without them the headline is 0.74 to 3.48. Counting each distinct sentence once per firm and year, AI lines per 100 duty lines are 4.1 in 2026 against 3.3 when every line counts.

Occupations. On thirds cut by the title-based occupation codes (Appendix A.3), which rest on the title rather than the duties, the high-use third goes from 0.77 to 3.28 and the low-use third is at 0.38 in 2026; on thirds of Eloundou et al.'s exposure, rated before the period, the high-exposure third goes from 0.67 to 2.15 and the low-exposure third is at 0.13. The high–low gap in genuinely new work widened by 0.09 points a year over 2019H2–2022; extended to 2026 that trend gives a gap of 0.8, against the 3.1 observed, 2.4 points above trend (1.5 on exposure thirds). In the occupation-level design, the exposure measure alone gives 1.77 points per standard deviation; the occupation's 2019–21 share of remote postings, entered alongside observed use, takes the gradient (+2.95 against −0.16), and its posting growth from 2019 to 2021 does not (1.60 for use).

Readings. Read in batches of one class and cell, genuinely new work in the high-use third is 0.77 in 2022 and 2.84 in 2026, against 0.74 and 3.48 on the mixed-batch readings; the difference is the AI lines, which the one-class batches read as new tasks less often (Appendix B.6). On Opus's readings of a fifth of the same batches, the figure is 0.52 in 2022 (0.04–1.01) and 3.08 in 2026 (2.42–3.74), against 3.48 (3.05–3.90) on Sonnet's, so the interval that covers either reader is 2.42–3.90 in 2026 and 0.04–1.01 in 2022; for all occupations in 2026 it is 1.27–1.91. The main estimate applies the new-task shares of 2023–24 to the new AI lines of 2019–22, pools the AI shares across thirds and gives governing and other AI lines the pooled share of the two sides. Read directly (in one-class batches), new AI lines of 2019–22 are new tasks 5% (building) and 14% (working with AI) of the time, governing and other AI lines 44%, and AI lines of the low-use third in 2026 31% and 69%; with those shares in place of the borrowed ones the headline is 0.61 to 3.34 in the high-use third and 0.26 in the low-use third. The screen misses new tasks of both kinds, and the second reader calls fewer non-AI lines new tasks: scaled by the recall rates of Table B1 and Appendix B and across the two readers and prompts, new non-AI tasks in the high-use third in 2026 lie between 0.8 and 2.8 per 100 duty lines and new AI tasks between 2.5 and 2.9; their rise since 2022 lies between 0.3 and 1.1 points for non-AI tasks against 2.4–2.7 for AI tasks.

Without the screen. Reading random duty lines of 2022 and 2026 in the high- and low-use thirds, flagged or not and in every band of similarity (3,000 lines per cell, Opus on 600), gives the

rates that the screen and the reading together would find if the screen missed nothing. In the high-use third in 2026, new AI tasks are 3.9 per 100 duty lines (standard error 0.34) and new non-AI tasks 2.3 (0.29) by Sonnet, and 3.6 and 1.5 by Opus; in 2022, they are 0.23 and 0.67 (Sonnet). In the low-use third in 2026 they are 0.17 and 0.29. The screened measure gives 2.51 and 0.97 for the high-use third in 2026, so the screen passes about two in three of the new AI tasks and four in ten of the non-AI ones, and part of the unscreened non-AI count is AI work that the keyword labels miss, such as building MCP servers or training frontier models. The ordering holds without the screen.

Screen. Table D1 gives the headline with the threshold set so that 5, 7.5, 12.5, 15 or 20% of 2019H2 duty lines count as new, each with the posting panel, reading samples and estimate rebuilt at that threshold, and with the January 2019 crawl dropped from the reference (Appendix B). The levels move with the threshold and the contrast between the thirds does not.

Drafted adverts and boilerplate. Benefits and legal text, which no reading counts as work, is flagged new 10% of the time in 2019H2 and 34% in 2026 (duty lines: 10% and 17.4%), the drafting change that the screen picks up; read with the paper's prompt, 99% of 800 such flagged lines of 2022 and 2026 are not duties and none is a new task, by either model. Appendix C gives the recurrence of AI lines across firms.

Sectors on each basis. Section 4 compares sectors on every duty line with the classifier's labels. On that basis, in 2026, information firms have 6.3 building-AI and 2.3 working-with-AI lines per 100 duty lines, finance 2.1 and 1.3, health care 0.4 and 0.3, and retail 0.4 and 0.3; corrected for the keywords' catch rates, the corresponding rates are 8.9 and 2.6, 3.2 and 1.5, 0.6 and 0.3, and 0.5 and 0.4. The correction raises the building side everywhere and changes no ordering. Across the other exhibits, Figure 1 and the occupation design count every duty line with the classifier's labels and read the new ones; Figure 2, the sector comparison and the pay regressions count every duty line with the classifier's labels; Figure 4b counts keyword lines; and the overall share of working with AI applies the catch-rate correction.

Pay. Table D2 gives the pay regressions with their identifying samples. Within firm, title and year, 1,751 postings in 500 cells identify the building-AI premium and 880 in 270 the working-with-AI premium, of 428,836 postings with pay. The AI indicators miss AI duties that have no keyword and that the classifier does not find, so some comparison postings carry the duty; under classical misclassification with precision 0.98 and sensitivity 0.60 (building) and 0.88 (working with AI), the estimates understate the premium by a factor that depends on the share of flagged postings in the identifying sample, and dividing by it gives 0.104 (0.006) and 0.024 (0.008) within firm, occupation and year and 0.015 (0.012) and 0.024 (0.015) within firm, title and year. In the firms that posted in every year 2022–26 (185,340 postings with pay), the premia within firm, occupation and year are 0.065 (0.006) and 0.017 (0.010), and within firm, title and year −0.008 (0.009) and 0.047 (0.027).

Work new to the job, by activity. Appendix B.2 explains why matching lines to O*NET's work activities does not give a reliable task for each line. A looser version is still informative about change over time. Opus 5.5 labelled 14,664 duty lines of 2019–26, drawn by year with AI lines over-sampled, with one of O*NET's 313 intermediate work activities, or none; a classifier on the sentence

Table D2: Posted-pay premia: specifications, identifying samples and variants

Fixed effects	Building AI	Working with AI	Postings	Build: switching postings / cells	Work: switching postings / cells
All postings with pay, 2023–26					
Occupation × state × year	0.170 (0.012)	0.054 (0.008)	428,836	171,739 / 2,566	159,919 / 2,114
Occupation × state × year × seniority	0.148 (0.011)	0.023 (0.007)	428,836	137,913 / 4,223	123,137 / 3,328
Firm × occupation × year	0.068 (0.004)	0.022 (0.008)	428,836	42,665 / 6,247	26,335 / 3,958
Firm × occupation × seniority × state × year	0.045 (0.004)	0.007 (0.010)	428,836	20,839 / 4,632	10,665 / 2,494
Firm × title × year	0.009 (0.007)	0.022 (0.014)	428,836	1,751 / 500	880 / 270
AI indicators from keyword lines only					
Firm × occupation × year	0.065 (0.005)	0.024 (0.008)	428,836	34,458 / 4,986	23,787 / 3,596
Firm × title × year	0.003 (0.010)	0.021 (0.015)	428,836	1,308 / 358	813 / 246
States and dates with a pay-range law					
Firm × occupation × year	0.062 (0.005)	0.011 (0.010)	192,822	19,050 / 3,097	11,564 / 1,892
Firm × title × year	0.009 (0.013)	0.056 (0.035)	192,822	677 / 209	289 / 91
Bottom of the posted range					
Firm × occupation × year	0.066 (0.007)	0.017 (0.010)	428,836	42,665 / 6,247	26,335 / 3,958
Firm × title × year	0.025 (0.016)	−0.013 (0.028)	428,836	1,751 / 500	880 / 270
Top of the posted range					
Firm × occupation × year	0.069 (0.004)	0.024 (0.008)	428,836	42,665 / 6,247	26,335 / 3,958
Firm × title × year	0.007 (0.007)	0.027 (0.015)	428,836	1,751 / 500	880 / 270
Firms posting in every year 2022–26					
Occupation × state × year	0.174 (0.022)	0.062 (0.011)	185,340	65,319 / 1,516	58,316 / 1,216
Firm × occupation × year	0.065 (0.006)	0.017 (0.010)	185,340	24,651 / 2,953	16,274 / 2,031
Firm × title × year	−0.008 (0.009)	0.047 (0.027)	185,340	1,064 / 297	429 / 128
Corrected for the recall of the AI labels					
Firm × occupation × year	0.104 (0.006)	0.024 (0.008)	428,836	factor 0.66	factor 0.94
Firm × title × year	0.015 (0.012)	0.024 (0.015)	428,836	factor 0.58	factor 0.90

Notes: Log annual posted pay, 2023–26, on indicators for a building-AI and a working-with-AI duty, with the fixed effects in the first column; standard errors clustered by firm in parentheses. Switching postings / cells: postings, and fixed-effect cells, in which the indicator varies. Keyword lines only: indicators from Sonnet's census of AI-keyword sentences, without the classifier. Corrected: the estimate and standard error divided by the misclassification factor in the last columns.

embeddings learns the labels (held out, it gives Opus's activity for 40% of lines and has it among its three likeliest for 64%) and scores every duty sentence. Each occupation's activity boundary is the set of activities that make up at least half a per cent of its duty lines in a random half of its 2019 postings, and the crossover is the share of its duty lines in a year that fall outside that boundary; the other half of 2019 is the placebo, what sampling and classifier noise alone produce. Over the 69 occupations with at least 2,000 such lines, the crossover is 14.0% in the 2019 placebo half, 15.4% in 2022 and 15.2% in 2026, 14.4% in the high-use third and 16.3% in the low-use third in 2026 (13.8% and 13.4% in the placebo), and the same without Workday and at boundaries of a quarter or one per cent. Opus's rewordings of 1,000 2019 lines keep the classifier's activity for 65% of lines and cross their occupation's boundary 27.9% of the time, against 27.1% for the originals, so wording alone moves the measure by less than a point. AI lines are 2.8% of the crossing lines of 2025–26 against 4.1% of all lines: the new AI duties mostly sit inside activities the occupation already had. On this measure, jobs have not moved into activities outside their 2019 mix since 2022.

Tools in every O*NET family. Extending Section 4.4 to all 48 O*NET 27.1 software categories with hot technologies, grouped into 16 families with 179 terms, with Opus 5.5 checking 80 matched lines in each cell as before, the share of postings asking for a family their occupation did not use in 2019–20 rose by 0.88 points between 2022 and 2026 for the five families of Section 4.4 (0.04 over 2019–22), by 0.60 for the eleven other families (0.31 over 2019–22, mostly collaboration tools in 2020–22) and by 1.38 for all sixteen. Figure D1 shows each family's change with its interval. Business intelligence (0.42) and workflow automation (0.31) move most; business systems (0.13), office software (0.13) and cloud and development tools (0.11) rise by about a tenth of a point; operating systems, web development, spreadsheets and computer-aided design do not move. Most families rise a little, by a median of 0.05 points: a general increase in the naming of tools as

adverts lengthen, which the two data families of Section 4.4 exceed several times over, and which the placebo of Appendix B.2 (non-use defined on 2019, new tools counted in 2020–21) does not show before 2022. The precision of the matches is low where an occupation did not use cloud, social-media or web tools, so those families lean on the correction.

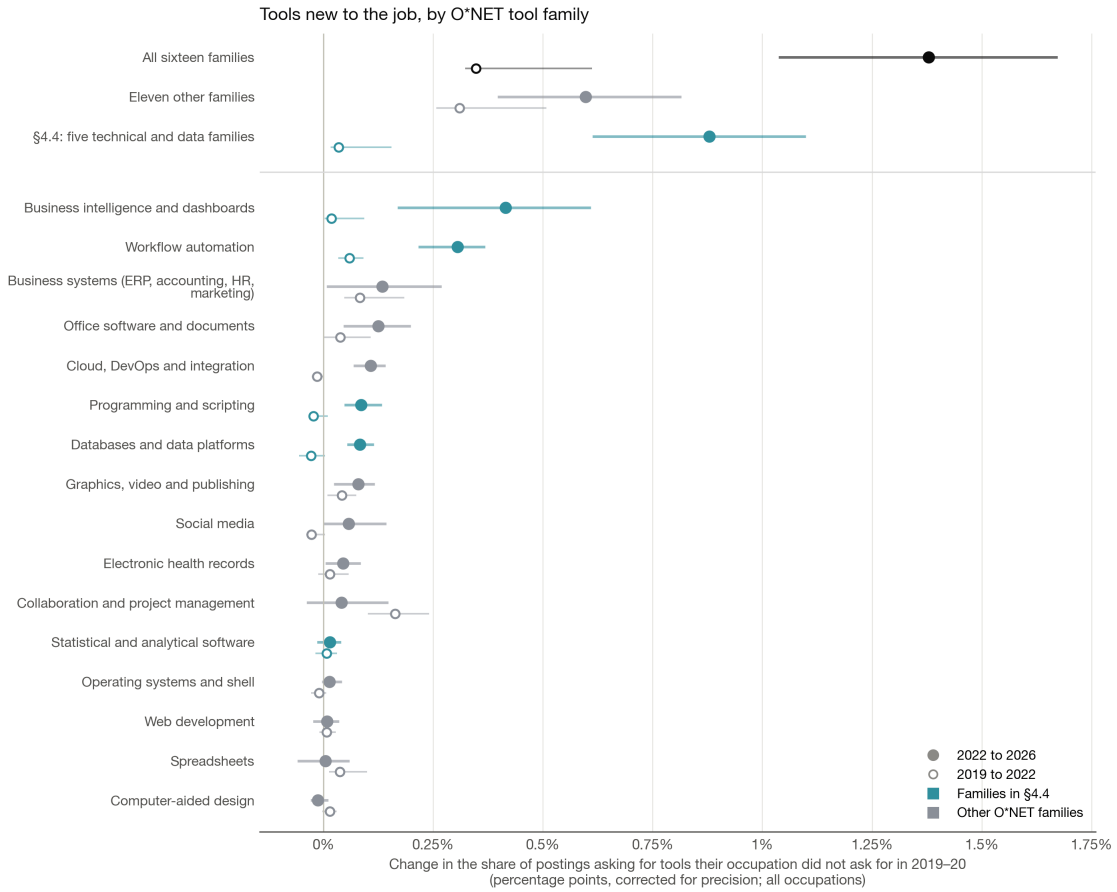


Figure D1: Tools new to the job, by O*NET tool family: change in the share of postings asking for a family their occupation did not use in 2019–20, corrected for the precision of the matches, 2022 to 2026 (filled) and 2019 to 2022 (open), with 95% intervals from resampling employers and the validation sample; all occupations. Teal: the five families of Section 4.4; grey: the eleven other families of O*NET 27.1's hot technologies; black: all sixteen.